

Quantum Speedup for Markov's Chains Mixing Time: Analysis and Implementation

Francesca Fino, Carlo Maggi ¹

June 2026

¹Supported by Prof. Nicolas Macris.

Markov chains are fundamental stochastic processes characterized by the memoryless property, where the probability of transitioning to a future state depends solely on the current state. They provide a powerful framework for sampling from complex probability distributions, even when the underlying normalization constant is unknown. In such applications, the efficiency of the process is governed by the chain's mixing time, which therefore serves as a natural measure of performance. Quantum algorithms, on the other hand, have demonstrated the potential to solve certain problems exponentially or quadratically faster than classical methods. Within this context, the prospect of reducing the mixing time of Markov chains using quantum algorithms becomes particularly compelling. In this work, we analyze the algorithms proposed in [1] and provide a comprehensive and reformulated exposition of its core theoretical framework, alongside an independent implementation designed to validate and analyze the behavior of non-reversible quantum walks.

1 Quantum Speedup for Nonreversible Markov Chains

Quantum algorithms can accelerate sampling from the stationary distribution of classical Markov chains, this is typically achieved via Szegedy’s quantum walk, which produces a unitary whose spectral gap is quadratically larger than the classical chain, resulting in a quadratic speedup. Many practical applications, however, involve nonreversible dynamics, where standard reversible techniques cannot be applied directly. The work we studied ([1]) extends quantum speedup techniques to nonreversible Markov chains by constructing quantum operations that approximate reflections through the stationary distribution, enabling accelerated sampling even in the nonreversible setting.

The authors propose two methods to construct an approximate reflection through the coherent stationary state. Both methods begin by performing a small number of classical steps until the chain is “reversible enough”, then apply a suitable quantum singular-value or eigenvalue transform to produce the reflection. In our report, we focus on the first method, while briefly explaining the second one.

A First method

The first method for accelerating nonreversible Markov chains begins by constructing a quantum encoding of the Markov kernel P and its time-reversal $P^\star$, capturing both the forward and backward dynamics of the chain:

$$\begin{aligned}\square &= \square_P = \sum_{x,y \in \mathbb{S}} \sqrt{P(x,y)} |x\rangle |y\rangle \langle x| \\ \square^\star &= \square_{P^\star} = \sum_{x,y \in \mathbb{S}} \sqrt{P^\star(x,y)} |x\rangle |y\rangle \langle x|\end{aligned}$$

From these encodings, we build a Projected Unitary Encoding $(\text{SWAP}, \square^\star, \square)$ of the curved discriminant $\mathcal{D}$:

$$\begin{aligned}\mathcal{D} &= \sqrt{PP^\star} \\ \mathcal{D} &= \square^\star \text{SWAP} \square.\end{aligned}$$

Since P is nonreversible, the singular values of the curved discriminant $\mathcal{D} = \sqrt{PP^\star}$ are equal to the square root of the eigenvalues of the multiplicative reversibilization $P^\star P$. However, the spectral gap of $P^\star P$ may be much smaller than that of P , and directly applying GQSVT to $\mathcal{D}$ without modification would negate the expected quadratic speedup. To address this, we perform a number of classical reversibilization steps, denoted τ_{rev} , effectively replacing $\mathcal{D}$ with the discriminant corresponding to the iterated kernel. These steps increase the singular-value gap associated with the stationary distribution, ensuring that the leading singular vector is well separated from the others.

Then, a fast-forwarding polynomial v is applied to the singular values of $\mathcal{D}_{\tau_{rev}}$ to isolate the leading singular vector corresponding to the stationary distribution. The polynomial is defined as

$$v_{\varepsilon,k}(x) = \varepsilon T_k(T_{1/k}(1/\varepsilon)x), \tag{1}$$

where k is the degree of the polynomial

$$k = O\left(\frac{\log(1/\varepsilon)}{\sqrt{\delta}}\right), \quad (2)$$

δ is the singular-value gap of $\mathcal{D}_{\tau_{rev}}$, and ε is the desired approximation precision. This polynomial transformation is implemented using the Generalized Quantum Singular Value Transform (GQSVT) on $\mathcal{D}_{\tau_{rev}}$:

$$v(D_{\tau_{rev}}) \simeq |\pi\rangle\langle\pi|, \quad (3)$$

where $|\pi\rangle$ is the encoding of the stationary distribution defined as

$$|\pi\rangle = \sum_{x \in \mathbb{S}} \sqrt{\pi(x)} |x\rangle. \quad (4)$$

The complexity of this method is governed by two factors: the reversibilization time τ_{rev} and the degree of the polynomial k . More precisely, the polynomial has degree

$$k = O\left(\frac{\log(1/\varepsilon)}{\sqrt{\delta}}\right)$$

$$k = O\left(\frac{\log(1/\varepsilon)}{\sqrt{\gamma(P^{\tau_{rev}}(P^*)^{\tau_{rev}})}}\right)$$

Where the equality comes from the fact that the singular values of $\mathcal{D}_{\tau_{rev}}$ are equal to the square root of the eigenvalues of $P^{\tau_{rev}}(P^*)^{\tau_{rev}}$. Then, using Taylor expansion, the spectral gaps are related as follows:

$$\begin{aligned} \delta &= 1 - \sigma_2(\mathcal{D}_k) \\ &= 1 - \sqrt{\lambda_2(P^{\tau_{rev}}(P^*)^{\tau_{rev}})} \\ &= 1 - \sqrt{1 - \gamma(P^{\tau_{rev}}(P^*)^{\tau_{rev}})} \\ &\simeq 1 - 1 + \frac{\gamma(P^{\tau_{rev}}(P^*)^{\tau_{rev}})}{2} \\ &\simeq \gamma(P^{\tau_{rev}}(P^*)^{\tau_{rev}}) \end{aligned}$$

Taking into account the reversibilization, the final complexity is

$$\begin{aligned} O\left(\frac{\tau_{rev}}{\sqrt{\gamma(P^{\tau_{rev}}(P^*)^{\tau_{rev}})}} \log(1/\varepsilon)\right) &\stackrel{(a)}{=} O\left(\frac{\tau_{rev}}{\sqrt{\gamma_{\infty}(P)^{\tau_{rev}}}} \log(1/\varepsilon)\right) \\ &= O\left(\sqrt{\frac{\tau_{rev}}{\gamma_{\infty}(P)}} \log(1/\varepsilon)\right) \\ &\stackrel{(b)}{=} O(\sqrt{\tau_{rev}\tau_{\varepsilon}} \log(1/\varepsilon)) \end{aligned}$$

Where the equalities come from:

- a We select τ_{rev} to be the smallest integer $k \geq 1$ where the pseudo-spectral gap $\gamma_\infty(P)$ is achieved:

$$\gamma_\infty(P) = \max_{k \geq 1} \frac{\gamma(P^k(P^\star)^k)}{k}$$

$$\gamma_\infty(P) = \frac{\gamma(P^{\tau_{rev}}(P^\star)^{\tau_{rev}})}{\tau_{rev}}$$

Therefore,

$$\gamma(P^{\tau_{rev}}(P^\star)^{\tau_{rev}}) = \gamma_\infty(P)\tau_{rev} \quad (5)$$

- b For a general ergodic chain, the mixing time $\tau_\epsilon(P)$ is of the order $O(1/\gamma_\infty(P))$

In the next sections, we provide detailed explanations and definitions to understand precisely this method.

B Second method

The second method is designed for cases where the time-reversal $P^\star$ is not accessible, so the curved discriminant cannot be constructed. Instead, it uses the flat discriminant D , defined as

$$D(x, y) = \sqrt{P(x, y)P(y, x)},$$

which is symmetric and therefore has real eigenvalues.

Since P is nonreversible, the leading eigenvector of D , denoted $|\mu\rangle$, does not, in general, coincide with the stationary distribution $|\pi\rangle$. To formalize when the method is effective, the algorithm relies on the notion of reversibility on π -average, defined through the expectation

$$\langle \pi | D | \pi \rangle = \sum_{x, y \in \mathbb{S}} \pi(x) \sqrt{P(x, y)P(y, x)} \pi(y), \quad (6)$$

which quantifies how close the chain is to being reversible in expectation with respect to π . If this quantity satisfies

$$1 - \langle \pi | D | \pi \rangle \ll \gamma(Q), \quad (7)$$

where Q is the geometric reversibilization associated with $|\mu\rangle$ and $\gamma(Q)$ its spectral gap, then the overlap

$$|\langle \pi | \mu \rangle|^2 \geq 1 - \frac{\langle \pi | \pi \rangle - \langle \pi | D | \pi \rangle}{\gamma(Q)} \quad (8)$$

is close to 1, ensuring that $|\mu\rangle$ approximates $|\pi\rangle$ well.

The method first performs a number of classical steps of the chain until the reversibility-on- π -average condition is satisfied, producing a modified flat discriminant $D_{t_{rev}}$.

A polynomial transformation is then applied to the eigenvalues of $D_{t_{rev}}$. The applied polynomial is a composition of two parts:

- a leading-eigenvalue-selection polynomial that isolates the leading eigenvalue from the rest of the spectrum,

- the fast-forwarding polynomial from the first method, which amplifies the component corresponding to the stationary distribution (eigenvalue 1) while suppressing all other eigenvalues.

Since $D_{t_{\text{rev}}}$ is symmetric, the resulting transformation is implemented via Generalized Quantum Eigenvalue Transform (GQET), producing a Projected Unitary Encoding of the approximate projector onto $|\mu\rangle$:

$$p(D_{t_{\text{rev}}}) \simeq |\mu\rangle\langle\mu|.$$

The overall complexity of the method for the flat discriminant depends on both the number of classical steps t_{rev} required to satisfy the reversibility-on- π -average condition and the degree of the composite polynomial. Explicitly, the quantum runtime scales as

$$T_{\text{quantum}} = O\left(t_{\text{rev}} \cdot k\right) = O\left(t_{\text{rev}} \cdot \frac{\log(1/\varepsilon)}{\sqrt{\delta}}\right),$$

where k is the degree of the fast-forwarding polynomial, δ is the effective spectral gap of the modified flat discriminant $D_{t_{\text{rev}}}$, and ε is the desired approximation error. By properly selecting t_{rev} and the polynomial, the top eigenvector $|\mu\rangle$ remains closely aligned with the stationary distribution $|\pi\rangle$, achieving a quantum speedup comparable to the first method even without access to $P^\star$.

2 Projected unitary encoding (PUE)

Projected unitary encoding is a method which can be used to encode inside a quantum computer a non-unitary general matrix A of dimension $N_L \times N_R$.

Definition 2.1. Given a matrix A of dimensions $N_L \times N_R$, a left isometry $\square_L$ of dimension $N \times N_L$, a right isometry $\square_R$ of dimension $N \times N_R$ and a unitary matrix U of dimension $N \times N$ we say that $(U, \square_L, \square_R)$ is a PUE of A iff

$$\square_L^\dagger \times U \times \square_R = A \quad (9)$$

Since the goal is to implement the LHS of (9) in the quantum computer, we need it to be unitary, however it is a well known fact that all unitary matrices are isometries, but not all isometries are unitaries. So the question is: how can we transform an isometry into a unitary matrix so that it can be implemented by a quantum computer?

By answering the question we make the LHS of (9) implementable and thus possible to encode any non-unitary matrix A in a quantum computer.

A Unitary extension of an isometry

Let's first recall the definition of isometry

Definition 2.2 (Isometry). An isometry is a linear map

$$V : \mathbb{C}^n \rightarrow \mathbb{C}^m \quad (10)$$

that preserves the inner product

$$\langle Vx, Vy \rangle = \langle x, y \rangle \quad (11)$$

where x and y are 2 vectors living in $\mathbb{C}^n$

From this definition, the corollary that interests us the most is the following:

Corollary 2.3. *Given an isometry V , the following holds true:*

$$V^\dagger \times V = I_n \quad (12)$$

Proof. From the following adjoint property

$$\langle Va, b \rangle = \langle a, V^\dagger b \rangle \quad (13)$$

we get that

$$\langle Vx, Vy \rangle = \langle x, V^\dagger Vy \rangle \quad (14)$$

thanks to the substitution

$$a = x, b = Vy \quad (15)$$

This leads us to

$$\langle Vx, Vy \rangle = \langle x, y \rangle = \langle x, V^\dagger Vy \rangle \quad (16)$$

which implies thanks to the linearity of the inner product that

$$y = V^\dagger Vy \implies y - V^\dagger Vy = 0 \implies V^\dagger V = I \quad (17)$$

□

The goal now is to construct a unitary matrix that encode the isometry. To do that we use the following facts:

1. Columns of V form an orthonormal basis of $\mathbb{C}^n$.
2. A square matrix where both rows and columns form an orthonormal basis is a unitary matrix.
3. $m \geq n$

The first fact follows from (12), the second fact is a well known linear algebra notion and we leave to the reader the proof, the third fact is a consequence of the first fact.

Definition 2.4 (Unitary extension). The unitary extension of isometry V is a matrix $U \in \mathbb{C}^m \times \mathbb{C}^m$ such that

$$U = \left(\begin{array}{ccc|ccc} | & & | & | & & | \\ v_1 & \cdots & v_n & w_{n+1} & \cdots & w_m \\ | & & | & | & & | \end{array} \right), \quad (18)$$

where $v_1, \dots, v_n$ are the columns of V , and $w_{n+1}, \dots, w_m$ are obtained by completing the orthonormal set $\{v_1, \dots, v_n\}$ to an orthonormal basis of $\mathbb{C}^m$, for example using the Gram–Schmidt process.

Now, since U is square and it has orthonormal columns, it also has orthonormal rows. Therefore, by fact 2, we conclude that U is unitary.

To simulate the isometry $V \in \mathbb{C}^{m \times n}$ on a quantum computer, we proceed as follows. First, we find the smallest number of qubits x such that

$$2^x \geq n.$$

This means that the input space $\mathbb{C}^n$ can be embedded into the computational basis of x qubits, whose Hilbert space has dimension 2^x . If n is not a power of two, we pad the remaining basis states with zero amplitudes.

Similarly, we choose the smallest number of qubits y such that

$$2^y \geq m.$$

This allows us to embed the output space $\mathbb{C}^m$ into the Hilbert space of y qubits.

In order to implement the isometry V , we construct a unitary extension U acting on a larger Hilbert space. We add $y - x$ ancillary qubits initialized in the state $|0\rangle$, so that an input state

$$|\psi\rangle \in \mathbb{C}^{2^x}$$

is embedded as

$$|\psi\rangle |0\rangle^{\otimes(y-x)} \in \mathbb{C}^{2^y}.$$

The unitary U is then constructed such that

$$U \left(|\psi\rangle |0\rangle^{\otimes(y-x)} \right) = V |\psi\rangle$$

on the subspace corresponding to the original input space.

3 Symmetric Projected Unitary Encoding (SPUE)

We now consider a special case of projected unitary encoding where the operator to encode is square and the encoding can be made symmetric. Let $A \in \mathbb{C}^{N \times N}$ be a square matrix. Suppose we have a symmetric unitary U ($U = U^\dagger$) and a single isometry $\square$ such that $N_L = N_R = N$ and $\square_L = \square_R = \square$.

Definition 3.1 (SPUE). The pair $(U, \square)$ is called a symmetric projected unitary encoding (SPUE) of A if

$$\square^\dagger U \square = A, \tag{19}$$

where $\square$ is an isometry ($\square^\dagger \square = I_N$) and U is symmetric ($U = U^\dagger$).

4 Qubitization

To apply quantum singular value transforms (QSVT) or quantum eigenvalue transforms (QGET), the operator of interest must be unitary. Classical Markov kernels and their discriminants are generally non-unitary, preventing them from being used directly in these quantum algorithms. Qubitization embeds a non-unitary operator A into a larger unitary walk operator $\mathcal{W}$ using a symmetric projected unitary encoding (SPUE) $(U, \square)$. This construction preserves the spectral information: each eigenvalue λ of A maps to a pair of eigenvalues $e^{\pm i\theta}$ of $\mathcal{W}$, with $\theta = \arccos(\lambda)$. In this way, polynomial transformations applied to $\mathcal{W}$ correspond to transformations on A . In the following, we formalize this construction.

The qubitized walk operator $\mathcal{W}$ is defined from a SPUE $(U, \square)$ of A .

Definition 4.1 (Qubitized walk operator). Let $(U, \square)$ be a symmetric projected unitary encoding. Denote by $R_\square$ the reflection operator about the range of $\square$,

$$R_\square = 2\square\square^\dagger - 1. \quad (20)$$

The qubitized walk operator $\mathcal{W}$ is defined by

$$\mathcal{W} = R_\square U \quad (21)$$

The subsequent proposition explicitly shows how $\mathcal{W}$ encodes the spectral properties of A and how its eigenvectors can be constructed from those of A .

Proposition 4.2. *To each eigenvector-eigenvalue pair $(\lambda, |v\rangle)$ of A , there correspond eigenvector-eigenvalue pairs of $\mathcal{W}$. Define $\theta = \cos^{-1}(\lambda) \in [0, \pi]$. If $|\lambda| < 1$, $\mathcal{W}$ has eigenvalues $e^{\pm i\theta}$ with eigenvectors $\mu_\pm$ given by:*

$$|\mu_\pm\rangle = \frac{1}{\sqrt{2}\sin(\theta)}(e^{\pm i\theta} - U)\square|v\rangle. \quad (22)$$

If $\lambda = \pm 1$, $\mathcal{W}$ has eigenvalues ± 1 , with corresponding eigenvector $|\mu\rangle = \square|v\rangle$, in the range of $\square$ and another one in the kernel of $\square$.

Proof. We want to prove that $\mathcal{W}$ has eigenvector $|\mu\rangle$ with eigenvalue $e^{\pm i\theta}$,

$$\mathcal{W}|\mu_\pm\rangle = e^{\pm i\theta}|\mu_\pm\rangle. \quad (23)$$

We start by assuming $|\lambda| < 1$.

$$\begin{aligned}
\mathcal{W}|\mu_{\pm}\rangle &\stackrel{(a)}{=} \frac{1}{\sqrt{2}\sin(\theta)} R_{\square}(e^{\pm i\theta} U \square |v\rangle - U^2 \square |v\rangle) \\
&\stackrel{(b)}{=} \frac{1}{\sqrt{2}\sin(\theta)} (e^{\pm i\theta} 2\square\square^{\dagger} U \square |v\rangle - e^{\pm i\theta} U \square |v\rangle - R_{\square} \square |v\rangle) \\
&\stackrel{(c)}{=} \frac{1}{\sqrt{2}\sin(\theta)} (e^{\pm i\theta} 2\square\lambda |v\rangle - e^{\pm i\theta} U \square |v\rangle - R_{\square} \square |v\rangle) \\
&\stackrel{(d)}{=} \frac{1}{\sqrt{2}\sin(\theta)} (e^{\pm i\theta} 2\square\lambda |v\rangle - e^{\pm i\theta} U \square |v\rangle - \square |v\rangle) \\
&\stackrel{(e)}{=} \frac{1}{\sqrt{2}\sin(\theta)} (e^{\pm i\theta} (e^{\pm i\theta} + e^{\mp i\theta}) - e^{\pm i\theta} U - 1) \square |v\rangle \\
&= \frac{e^{\pm i\theta}}{\sqrt{2}\sin(\theta)} (e^{\pm i\theta} + e^{\mp i\theta} - U - e^{\mp i\theta}) \square |v\rangle \\
&= \frac{e^{\pm i\theta}}{\sqrt{2}\sin(\theta)} (e^{\pm i\theta} - U) \square |v\rangle \\
&= e^{\pm i\theta} |\mu_{\pm}\rangle
\end{aligned}$$

Where the equalities follow from:

- a We expand it using the definition of $\mathcal{W}$ (21) and $|\mu_{\pm}\rangle$ (22).
- b We expand R (20), and use the identity $U^2 = I$
- c We use the fact that $(U, \square)$ is SPUE of A and that $|v\rangle$ is eigenvector of A with eigenvalue λ
- d We use the identity $R_{\square} \square = \square$

$$R_{\square} \square = (2\square\square^{\dagger} - 1)\square = 2\square - \square = \square \quad (24)$$

- e We use the definition $\theta = \cos^{-1}(\lambda)$,

$$\lambda = \cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \quad (25)$$

If $\lambda = \pm 1$, then $\theta = 0, \pi$. In this case, the vectors $U \square |v\rangle$ and $\square |v\rangle$ are both normalized, because U is unitary and $\square$ is an isometry. Moreover,

$$\langle \square v | U \square v \rangle = \langle v | \square^{\dagger} U \square |v\rangle = \langle v | A |v\rangle = \lambda.$$

If $\lambda = 1$, then two normalized vectors have inner product 1, hence

$$U \square |v\rangle = \square |v\rangle.$$

If $\lambda = -1$, then two normalized vectors have inner product -1 , hence

$$U \square |v\rangle = -\square |v\rangle.$$

Therefore, in both cases,

$$U\Box|v\rangle = \lambda\Box|v\rangle.$$

It follows that

$$\mathcal{W}\Box|v\rangle = R_{\Box}U\Box|v\rangle = \lambda R_{\Box}\Box|v\rangle = \lambda\Box|v\rangle.$$

Thus, when $\lambda = \pm 1$, the vector $\Box|v\rangle$ is an eigenvector of $\mathcal{W}$ with eigenvalue $\lambda = \pm 1$. $\square$

5 Encoding Markov Chains in Quantum Computers

A Markov Chains

We begin by recalling key notions and properties of Markov chains to specify the domain of the algorithm. In particular, we focus on finite ergodic Markov kernels, which guarantees convergence to a unique stationary distribution. This stationary distribution is the target distribution of the sampling problem.

We then explain how the information contained in the Markov kernel is encoded on a quantum computer via the flat and curved discriminants, and how these encodings are used to construct the qubitized walk operator.

We start by giving a precise definition of a Markov Kernel.

Definition 5.1 (Markov kernel). Let $\mathbb{S}$ be a finite set of size $n \in \mathbb{N}$. A Markov kernel is a function $P : \mathbb{S}^2 \rightarrow [0, 1]$ such that

$$\forall x \in \mathbb{S} : \sum_{y \in \mathbb{S}} P(x, y) = 1.$$

The notion of irreducibility captures whether it is possible to reach any state from any other, which is crucial for ensuring the long-term behavior of the chain is well-defined.

Definition 5.2 (Irreducibility). Let P be a Markov kernel on the state space $\mathbb{S}$. It is said to be irreducible if and only if

$$\forall x, y \in \mathbb{S}, \exists t \in \mathbb{N} \quad \text{such that} \quad P^t(x, y) > 0.$$

Ergodicity further strengthens this property by guaranteeing uniqueness and non-degeneracy of the stationary behavior, so that the chain does not exhibit periodic or transient dynamics.

Definition 5.3 (Ergodic kernel). A Markov kernel P on a finite state space $\mathbb{S}$ is called ergodic if it is irreducible and has a single non-degenerate eigenvalue on the unit circle.

Under ergodicity, the chain converges to a limiting distribution independent of the initial state; this limiting distribution is formalized as the stationary distribution.

Proposition 5.4 (Stationary distribution). *Let P be an ergodic Markov kernel. Then there exists a function $\pi : \mathbb{S} \rightarrow [0, 1]$, called the stationary distribution, such that*

$$\forall x, y \in \mathbb{S} : \lim_{t \rightarrow \infty} P^t(x, y) = \pi(y).$$

To quantify how quickly the chain approaches its stationary distribution, we introduce the concept of the ε -mixing time, which measures the number of steps required for the chain to be close to stationarity within a specified precision.

Definition 5.5. Let $0 < \varepsilon < 1$. Denote the ε -mixing time of a Markov chain P by

$$\tau_\varepsilon = \min_{t \in \mathbb{N}} \{t : \max_{x \in \mathbb{S}} d_{TV}(P^t(x, \cdot), \pi)\} \quad (26)$$

The adjoint operator provides a notion of time-reversal and is used to define the curved discriminant. Understanding its relationship with the original kernel is essential for analyzing spectral properties and convergence behavior.

Definition 5.6 (Adjoint Markov kernel). Define the adjoint $P^\star$ of P as its adjoint with respect to the scalar product induced by π . More precisely, for $f, g : \mathbb{S} \rightarrow \mathbb{C}$, define

$$\langle f, g \rangle_\pi = \sum_{x \in \mathbb{S}} \pi(x) f(x) \overline{g(x)}.$$

Then $P^\star$ is the unique operator such that

$$\forall f, g : \mathbb{S} \rightarrow \mathbb{C} : \langle f, Pg \rangle_\pi = \langle P^\star f, g \rangle_\pi.$$

Equivalently, when written as a Markov kernel,

$$P^\star(x, y) = \frac{\pi(y)P(y, x)}{\pi(x)}.$$

To formalize the concept of reversibility in a Markov chain, we recall the following definition. This notion is crucial for understanding the focus of the paper on non-reversible chains.

Definition 5.7 (Reversible Markov kernel). A Markov kernel P with stationary distribution π is called reversible if

$$\pi(x)P(x, y) = \pi(y)P(y, x), \quad \forall x, y \in \mathbb{S}.$$

Equivalently, in operator form, a reversible kernel satisfies

$$P = P^\star,$$

where $P^\star$ is the adjoint of P with respect to the inner product induced by π .

This condition ensures detailed balance, meaning that the probability flow between any two states is symmetric under the stationary distribution.

While the absolute spectral gap governs the convergence rate of reversible chains, and the pseudo-spectral gap plays the analogous role for nonreversible chains, it is important to note that classically the mixing time scales as the inverse of the gap (or pseudo-gap), not its square root. In particular:

Definition 5.8 (Absolute spectral gap). Define the absolute spectral gap δ of a reversible Markov kernel P by

$$\delta = 1 - \max_{\lambda \in \sigma(P) \setminus \{1\}} |\lambda|.$$

Definition 5.9 (Pseudo-spectral gap). Let P be an ergodic nonreversible Markov kernel with adjoint $P^\star$. The pseudo-spectral gap of P is defined as

$$\gamma_\infty(P) = \max_{k \geq 1} \frac{\gamma(P^k(P^\star)^k)}{k},$$

where $\gamma(\cdot)$ denotes the spectral gap of the associated reversible kernel.

The classical ε -mixing times then scale as

$$\begin{aligned}\tau_\varepsilon(P) &= O\left(\frac{\log(1/\varepsilon)}{\delta}\right) \quad \text{for reversible chains,} \\ \tau_\varepsilon(P) &= O\left(\frac{\log(1/\varepsilon)}{\gamma_\infty(P)}\right) \quad \text{for nonreversible chains.}\end{aligned}$$

Notice that in both cases, the scaling is proportional to the inverse of the (pseudo-)spectral gap, up to logarithmic factors, in contrast to the quantum case where the polynomial-based singular-value transformations achieve a quadratic improvement, scaling with the inverse square root of the (pseudo-)spectral gap, as shown in the first section.

B Discriminants

We now construct matrices that encode the transition structure in a way suitable for quantum algorithms. These discriminant matrices will later serve as the target for polynomial transformations using GQSVT or GQET.

Definition 5.10 (Discriminant). Let (P, Q) be Markov kernels on a finite state space $\mathbb{S}$. The associated discriminant D is defined by

$$\forall x, y \in \mathbb{S} : D(x, y) = \sqrt{P(x, y)Q(y, x)}.$$

The curved discriminant incorporates the adjoint operator, encoding the asymmetry of the chain while preserving spectral information crucial for constructing quantum walk operators.

Definition 5.11 (Curved discriminant). The curved discriminant of a Markov kernel P is the discriminant associated with the pair $(P, P^\star)$. Hence it is the matrix $\mathcal{D}$ defined by

$$\forall x, y \in \mathbb{S} : \mathcal{D}(x, y) = \sqrt{P(x, y)P^\star(y, x)}.$$

To relate the spectral properties of the Markov kernel to its quantum encoding, we first note that the curved discriminant $\mathcal{D}$ is similar to P , ensuring that they share the same spectrum.

Proposition 5.12. *The curved discriminant $\mathcal{D}$ is similar to P .*

Proof. Let $x, y \in \mathbb{S}$. Then,

$$\begin{aligned}\mathcal{D}(x, y) &= \sqrt{P(x, y)P^\star(y, x)} \\ &= \sqrt{P(x, y)\frac{\pi(x)}{\pi(y)}P(x, y)} \\ &= \sqrt{\frac{\pi(x)}{\pi(y)}}P(x, y) \\ &= (MPM^{-1})(x, y),\end{aligned}$$

where $M(w, z) = \delta_{w,z}\sqrt{\pi(w)}$ for all $w, z \in \mathbb{S}$. Thus $\mathcal{D} = MPM^{-1}$, and hence $\mathcal{D}$ is similar to P . $\square$

An essential feature for quantum sampling is that the leading singular vector of the curved discriminant corresponds to the stationary distribution of the Markov chain. This ensures that applying a polynomial filter to select the largest singular value correctly prepares the target state for amplitude amplification. We now provide a rigorous proposition and proof showing that both the left and right singular vectors of $\mathcal{D}$ coincide with the coherent stationary state $|\pi\rangle$.

Proposition 5.13. *Let P be an ergodic Markov kernel with stationary distribution π (row vector). Then*

$$\pi P = \pi.$$

Proof. By definition, the stationary distribution satisfies

$$\pi(y) = \sum_{x \in \mathbb{S}} \pi(x) P(x, y), \quad \forall y \in \mathbb{S}.$$

Writing this in vector notation using a row-vector π gives

$$\pi P = \pi,$$

which shows that π is a left eigenvector of P corresponding to eigenvalue 1. $\square$

Proposition 5.14. *Let $P^\star$ denote the adjoint of P with respect to the π -weighted inner product, and let the curved discriminant be*

$$\mathcal{D} = \sqrt{PP^\star}.$$

Then the leading right and left singular vectors of $\mathcal{D}$ are

$$|\pi\rangle = \sum_{x \in \mathbb{S}} \sqrt{\pi(x)} |x\rangle,$$

with singular value $\sigma_1 = 1$.

Proof. Let $\mathcal{D} = \sum_i \sigma_i |w_i\rangle\langle v_i|$ be the singular value decomposition of $\mathcal{D}$, where $|v_i\rangle$ and $|w_i\rangle$ are the right and left singular vectors, respectively.

Right singular vector. The right singular vectors satisfy

$$\mathcal{D}^\dagger \mathcal{D} |v_i\rangle = \sigma_i^2 |v_i\rangle.$$

Since $\mathcal{D}^\dagger \mathcal{D} = PP^\star$, we consider

$$|v_1\rangle = \sum_{x \in \mathbb{S}} \sqrt{\pi(x)} |x\rangle.$$

Then

$$P^\star |v_1\rangle = |v_1\rangle, \quad \text{and} \quad P |v_1\rangle = |v_1\rangle,$$

implying that $|v_1\rangle$ is the right singular vector corresponding to $\sigma_1 = 1$.

Left singular vector. By definition of the SVD, the left singular vector corresponding to σ_1 is

$$|w_1\rangle = \frac{1}{\sigma_1} \mathcal{D}|v_1\rangle = \mathcal{D}|v_1\rangle = |v_1\rangle,$$

since $\sigma_1 = 1$. Thus, the left singular vector coincides with the right singular vector and encodes the coherent stationary state. □

When the adjoint is unknown or difficult to implement, we can instead consider the flat discriminant, which symmetrizes the transition probabilities and has a leading singular vector corresponding to the most reversible distribution $|\mu\rangle$, which approximates the stationary distribution when the chain is reversible on π -average.

Definition 5.15 (Flat discriminant). Let P be a Markov transition matrix over a state space $\mathbb{S}$ of size $|\mathbb{S}| = n$. The flat discriminant of P is the matrix D defined by

$$\forall x, y \in \mathbb{S} : D(x, y) = \sqrt{P(x, y)P(y, x)}.$$

To formalize the target of quantum reflections when the adjoint operator is unavailable, we define the most reversible distribution, which is the leading singular vector of the flat discriminant.

Definition 5.16 (Most reversible distribution). Let D be the flat discriminant of a primitive ergodic Markov kernel P . The most reversible distribution $|\mu\rangle$ is the eigenvector of D corresponding to its largest eigenvalue:

$$|\mu\rangle = \arg \max_{\nu > 0} \langle \nu | D | \nu \rangle,$$

where the maximization is over all strictly positive probability distributions ν on the state space.

To ensure that the leading singular vector of the flat discriminant approximates the stationary distribution, we introduce the condition of reversibility on π -average, which measures how close the chain is to being reversible when averaged over the stationary distribution.

Definition 5.17 (Reversibility on π -average). Let P be a Markov kernel with stationary distribution π , and let D_j be the flat discriminant of P^j . We say that P^j is reversible on π -average if

$$1 - \langle \pi | D_j | \pi \rangle \ll \gamma(Q_j),$$

where Q_j is the geometric reversibilization of P^j and $\gamma(Q_j)$ its spectral gap. This condition ensures that the overlap between $|\pi\rangle$ and the most reversible distribution $|\mu\rangle$ is close to 1.

C Encoding Markov Chains

In order to implement quantum algorithms for sampling from Markov chains, we now describe how the transition information of a kernel is encoded into quantum operators. This includes partial isometries representing the Markov kernel, and the construction of the flat and curved discriminants, which are subsequently used to define the qubitized walk operators.

Markov kernel

For each Markov kernel $P : \mathbb{S} \rightarrow [0, 1]$, we define the partial isometry

$$\square_P = \sum_{x,y \in \mathbb{S}} \sqrt{P(x,y)} |x\rangle |y\rangle \langle x| \quad (27)$$

It has dimensions $|\mathbb{S}|^2 \times |\mathbb{S}|$ and satisfied $\square_P^\dagger \square_P = I$.

Probability distribution

In order to prepare a quantum state representing the stationary distribution π of a Markov chain, we encode π into a coherent quantum state:

$$|\pi\rangle = \sum_{x \in \mathbb{S}} \sqrt{\pi(x)} |x\rangle. \quad (28)$$

This state is a normalized superposition over the computational basis states, where the amplitude of each basis state $|x\rangle$ is given by the square root of the classical probability $\pi(x)$.

Curved Discriminant

Let S be the swap operator defined as

$$S = \sum_{x,y \in \mathbb{S}} |x,y\rangle \langle x,y|. \quad (29)$$

We show that

$$(S, \square^\star, \square) = (S, \square_{P^\star}, \square_P) \quad (30)$$

is the projected unitary encoding of the curved discriminant.

$$\begin{aligned} \langle i | \square^\star S \square | j \rangle &= \langle i | \left(\sum_{x,y} \sqrt{P^\star(x,y)} |x\rangle \langle x,y| \right) \left(\sum_{a,b} |b,a\rangle \langle a,b| \right) \left(\sum_{x',y'} \sqrt{P(x',y')} |x',y'\rangle \langle x'| \right) | j \rangle \\ &= \langle i | \left(\sum_{x,y,x',y'} \sqrt{P^\star(x,y)P(x',y')} |x\rangle \langle y,x|x',y'\rangle \langle x'| \right) | j \rangle \\ &= \langle i | \left(\sum_{x,y} \sqrt{P^\star(x,y)P(y,x)} |x\rangle \langle y| \right) | j \rangle \\ &= \sqrt{P^\star(i,j)P(j,i)} \end{aligned}$$

Flat Discriminant

Setting $\square^\star = \square$, we show that $(S, \square)$ is the symmetric projected unitary encoding of the flat discriminant.

$$\langle i | \square S \square | j \rangle = \sqrt{P(i,j)P(j,i)} \quad (31)$$

Walk Operators

From these projected encodings, we construct the qubitized walk operator of the curved discriminant as:

$$\mathcal{W} = (2\Box\Box^\dagger - 1)S \quad (32)$$

and the qubitized walk operator of the flat discriminant as:

$$\mathcal{W} = R_{\bar{\Box}}S, \quad R_{\bar{\Box}} = |0\rangle\langle 0| \otimes R_{\Box_L} + |1\rangle\langle 1| \otimes R_{\Box_R} \quad (33)$$

In the last case, we used the notion of hermitianization which is going to be introduced in Section. A.

6 Chebyshev Polynomials

In this quantum algorithm, the goal is to prepare the stationary distribution of a Markov chain by transforming the spectra of the discriminant.

In the case of the curved discriminant, the leading singular value is exactly 1, corresponding to the stationary coherent state. This allows us to directly apply a Chebyshev fast-forwarding polynomial to isolate this component while suppressing all other singular values.

For the flat discriminant, the leading singular value may be less than 1. In this case, the fast-forwarding polynomial alone is insufficient, and one must compose it with a leading-eigenvalue selection polynomial to achieve the desired amplification of the stationary component.

A Fast-forwarding Polynomial

The Chebyshev fast-forwarding polynomial is the unique polynomial of minimal degree that, when applied to the discriminant, keeps the leading singular value equal to 1 while suppressing all other singular values within a specified error ε .

Theorem 6.1. *Let $0 < \varepsilon < 1$. Let $(\delta_k)_{k \in \mathbb{N}}$ be the sequence of positive real numbers defined, for each $k \in \mathbb{N}$, by*

$$\delta_k = \frac{T_{1/k}(1/\varepsilon) - 1}{T_{1/k}(1/\varepsilon)},$$

where $T_y(x) = \cosh(y \operatorname{arcosh}(x))$, $y \in (0, 1)$. Let $C(\varepsilon, k)$ be the set of real polynomials v of defined parity, having all their roots in $(-1, 1)$, and satisfying

- $v(1) = 1$,
- $|v(x)| \leq \varepsilon$, for all $x \in [-1 + \delta_k, 1 - \delta_k]$.

Then, there is no polynomial of degree less than k in $C(\varepsilon, k)$. Moreover, the polynomial $v_{\varepsilon, k}(x) = \varepsilon T_k(T_{1/k}(1/\varepsilon)x)$ satisfies $\max v_{\varepsilon, k}^{-1}(\varepsilon) = 1 - \delta_k$, and is the unique polynomial of degree k in $C(\varepsilon, k)$.

B Degree of the fast-forwarding polynomial

We now estimate the degree required for the Chebyshev fast-forwarding polynomial. The degree scales as $O(\delta^{-1/2} \log(1/\varepsilon))$, where δ is the spectral gap between the leading and next-largest singular value, and ε is the desired suppression of the non-leading components.

Proposition 6.2. *Assume that the target eigenvalue or singular value is equal to 1, and that all other eigenvalues or singular values lie in $[-1 + \delta, 1 - \delta]$, for some $0 < \delta < 1$. The fast-forwarding polynomial separates the leading eigenvalue or singular value at 1 from the rest of the spectrum using a degree that scales as*

$$O\left(\frac{\log(\frac{1}{\varepsilon})}{\sqrt{\delta}}\right). \tag{34}$$

This square-root dependence on the inverse gap explains the term “fast-forwarding”: the polynomial reproduces the spectral filtering effect of ordinary powers of the discriminant, but with degree $O(\delta^{-1/2} \log(1/\varepsilon))$ instead of $O(\delta^{-1} \log(1/\varepsilon))$.

Proof. We start by approximating the transition width of the polynomial proposed in Theorem 6.1,

$$\begin{aligned}\delta_k &= \frac{T_{1/k}(1/\varepsilon) - 1}{T_{1/k}(1/\varepsilon)} = 1 - \frac{1}{T_{1/k}(1/\varepsilon)} \\ &\stackrel{(a)}{=} 1 - \frac{1}{\cosh\left(\frac{L}{k}\right)} \\ &\stackrel{(b)}{=} \frac{L^2}{2k^2} + O\left(\frac{L^4}{k^4}\right)\end{aligned}$$

Where the equalities come from:

a We use the definition $T_y(x) = \cosh(y \operatorname{arcosh}(x))$, and we obtain

$$T_{1/k}(1/\varepsilon) = \cosh\left(\frac{L}{k}\right), \quad L = \operatorname{arcosh}(1/\varepsilon) \quad (35)$$

b For large k , we use the Taylor expansion

$$\cosh(t) = 1 + \frac{t^2}{2} + O(t^4), \quad t \rightarrow 0. \quad (36)$$

to approximate the expression

$$1 - \frac{1}{\cosh(t)} = 1 - \left(1 - \frac{t^2}{2}\right) + O(t^4) = \frac{t^2}{2} + O(t^4). \quad (37)$$

We take $t = \frac{L}{k}$ and obtain

$$1 - \frac{1}{\cosh\left(\frac{L}{k}\right)} = \frac{L^2}{2k^2} + O\left(\frac{L^4}{k^4}\right) \quad (38)$$

Then, we express the degree of the polynomial in terms of the spectral gap

$$\begin{aligned}k &= \frac{L}{\sqrt{2\delta_k}} + O(1) \\ &= \frac{\operatorname{arcosh}(1/\varepsilon)}{\sqrt{2\delta_k}} + O(1) \\ &\stackrel{(a)}{=} O\left(\frac{\log\left(\frac{1}{\varepsilon}\right)}{\sqrt{\delta_k}}\right)\end{aligned}$$

Where the equality comes from:

a We approximate $\operatorname{arcosh}(\frac{1}{\varepsilon}) = \log\left(\frac{1}{\varepsilon} + \sqrt{\frac{1}{\varepsilon^2} - 1}\right) = O(\log(\frac{1}{\varepsilon}))$

To suppress the whole non-leading spectrum $[-1 + \delta, 1 - \delta]$, we choose k such that $\delta_k \leq \delta$. Then

$$[-1 + \delta, 1 - \delta] \subseteq [-1 + \delta_k, 1 - \delta_k],$$

so Theorem 6.1 implies

$$|v_{\varepsilon,k}(x)| \leq \varepsilon \quad \text{for all } x \in [-1 + \delta, 1 - \delta].$$

Therefore,

$$k = O\left(\frac{\log(\frac{1}{\varepsilon})}{\sqrt{\delta}}\right).$$

□

This square-root scaling of the required polynomial degree, $O(\delta^{-1/2} \log(1/\varepsilon))$, in contrast to the classical scaling $O(\delta^{-1} \log(1/\varepsilon))$, is exactly where the quantum algorithm achieves its quadratic speedup over classical approaches.

C Scaling Factor

Let $v(x) = \sum_{n=0}^d a_n T_n(x)$ be a complex polynomial expressed in the Chebyshev basis, and let $\Upsilon(z) = \sum_{n=0}^d a_n z^n$ be its associated signal-processing polynomial.

In quantum signal processing, Υ must satisfy

$$\max_{z \in \partial B_1(0)} |\Upsilon(z)| \leq 1,$$

in order to be implemented directly. If this condition is not satisfied, we need to rescale the polynomial. This motivates the following definition.

Definition 6.3 (Scaling factor). Let v be a complex polynomial written in the Chebyshev basis, with associated signal-processing polynomial Υ . The *scaling factor* of v is

$$\beta = \frac{\max_{z \in \partial B_1(0)} |\Upsilon(z)|}{\max_{x \in [-1,1]} |v(x)|}.$$

Equivalently, β is the factor by which v must be divided so that the corresponding signal-processing polynomial satisfies the unitarity constraint on the unit circle. Implementing v/β instead of v reduces the success probability by a factor $1/\beta^2$.

For the fast-forwarding polynomial used in this work, no rescaling is needed. Recall

$$v_{\varepsilon,d}(x) = \varepsilon T_d(x T_{1/d}(1/\varepsilon)), \quad 0 < \varepsilon \leq 1.$$

By construction, $v_{\varepsilon,d}(1) = 1$, since

$$v_{\varepsilon,d}(1) = \sum_{n=0}^d a_n T_n(1) = \sum_{n=0}^d a_n = 1.$$

Moreover, evaluating the associated signal-processing polynomial on the unit circle gives

$$|\Upsilon(z)| = \left| \sum_{n=0}^d a_n z^n \right| \leq \sum_{n=0}^d |a_n| |z|^n = \sum_{n=0}^d a_n = 1,$$

where the inequality follows from the triangle inequality, $|z| = 1$, and the fact that all coefficients a_n of the fast-forwarding polynomial are non-negative. Therefore, the scaling factor is $\beta = 1$, and no additional normalization is required.

7 Generalized Quantum Signal Processing (GQSP)

Generalized Quantum Signal Processing (GQSP), [2], is a framework to implement a polynomial transformation of a unitary operator on a quantum state. More precisely, given a unitary U and a target polynomial P , GQSP implements the action

$$P(U) |\psi\rangle \quad (39)$$

on an input state $|\psi\rangle$, up to constraints on the polynomial's degree and amplitude.

To do this, we introduce an ancillary qubit and operate in the Hilbert space $\mathbb{C}^2 \otimes \mathbb{H}$, where $\mathbb{H}$ is the Hilbert space of U . We then define the following two unitary operators acting on the full space:

$$A = |0\rangle\langle 0| \otimes U + |1\rangle\langle 1| \otimes I = \begin{pmatrix} U & 0 \\ 0 & I \end{pmatrix}, \quad (40)$$

$$R(\theta, \phi, \lambda) = \begin{bmatrix} e^{i(\lambda+\phi)} \cos(\theta) & e^{i\phi} \sin(\theta) \\ e^{i\lambda} \sin(\theta) & -\cos(\theta) \end{bmatrix} \otimes I. \quad (41)$$

A Block notation

Given the basis $\{|0\rangle, |1\rangle\}$ of $\mathbb{C}^2$, any vector in $\mathbb{C}^2 \otimes \mathbb{H}$ can be written as

$$|0\rangle \otimes |\psi_0\rangle + |1\rangle \otimes |\psi_1\rangle = \begin{pmatrix} |\psi_0\rangle \\ |\psi_1\rangle \end{pmatrix} \quad (42)$$

Thus, any linear map on $\mathbb{C}^2 \otimes \mathbb{H}$ can be expressed as a 2×2 *block matrix*:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} |\psi_0\rangle \\ |\psi_1\rangle \end{pmatrix} = \begin{pmatrix} A|\psi_0\rangle + B|\psi_1\rangle \\ C|\psi_0\rangle + D|\psi_1\rangle \end{pmatrix} \quad (43)$$

B Encoding the polynomial

Theorem 7.1 (Generalized Quantum Signal Processing). $\forall d \in \mathbb{N}, \exists \vec{\theta}, \vec{\phi} \in \mathbb{R}^{d+1}, \lambda \in \mathbb{R} \text{ s.t.}$

$$\left(\prod_{j=1}^d R(\theta_j, \phi_j, 0) A \right) R(\theta_0, \phi_0, \lambda) = \begin{bmatrix} P(U) & \cdot \\ Q(U) & \cdot \end{bmatrix} = G \quad (44)$$

$\Longleftrightarrow$

1. $P, Q \in \mathbb{C}[x]$ and $\deg(P), \deg(Q) \leq d$.

2. $\forall x \in \mathbb{T}, |P(x)|^2 + |Q(x)|^2 = 1$.

Here $\mathbb{T} = \{x \in \mathbb{C} : |x| = 1\}$ is the complex unit circle.

Proof. The proof is done by induction on both sides.

($\Rightarrow$)

Base case $d = 0$.

$$G_0 = R(\theta_0, \phi_0, \lambda) = \begin{bmatrix} e^{i(\lambda+\phi_0)} \cos(\theta_0) & e^{i\phi} \sin(\theta_0) \\ e^{i\lambda} \sin(\theta_0) & -\cos(\theta_0) \end{bmatrix} \quad (45)$$

We immediately see that 1 and 2 hold, in fact we get that

$$P = e^{i(\lambda+\phi_0)} \cos(\theta_0) \quad (46)$$

$$Q = e^{i\lambda} \sin(\theta_0) \quad (47)$$

Since P and Q are constants their degree is 0, thus condition 1 is satisfied. And also $|P|^2 + |Q|^2 = \cos(\theta_0)^2 + \sin(\theta_0)^2 = 1$, hence condition 2 is also satisfied.

Induction step

Assume that for degree $d - 1$ we already have

$$G_{d-1} = \begin{pmatrix} \widehat{P}(U) & \cdot \\ \widehat{Q}(U) & \cdot \end{pmatrix}$$

$$\deg(\widehat{P}), \deg(\widehat{Q}) \leq d - 1$$

$$|\widehat{P}|^2 + |\widehat{Q}|^2 = 1$$

Then for degree d according to (44)

$$\begin{aligned} G_d &= R(\theta_d, \phi_d, 0) A G_{d-1} \\ &= \begin{bmatrix} e^{i(\phi_d)} \cos(\theta_d) & e^{i\phi_d} \sin(\theta_d) \\ \sin(\theta_d) & -\cos(\theta_d) \end{bmatrix} \begin{pmatrix} U & 0 \\ 0 & I \end{pmatrix} \begin{pmatrix} \widehat{P}(U) & \cdot \\ \widehat{Q}(U) & \cdot \end{pmatrix} \\ &= \begin{bmatrix} e^{i(\phi_d)} \cos(\theta_d) U \widehat{P}(U) + e^{i\phi_d} \sin(\theta_d) \widehat{Q}(U) & \cdot \\ \sin(\theta_d) U \widehat{P}(U) - \cos(\theta_d) \widehat{Q}(U) & \cdot \end{bmatrix} \end{aligned}$$

From which we get that

$$\begin{aligned} P(U) &= e^{i(\phi_d)} (\cos(\theta_d) U \widehat{P}(U) + \sin(\theta_d) \widehat{Q}(U)) \\ Q(U) &= \sin(\theta_d) U \widehat{P}(U) - \cos(\theta_d) \widehat{Q}(U) \end{aligned}$$

From these two expressions it is immediate to see that condition 1 is satisfied. While condition 2 follows from the fact that G_d is a product of unitary matrices and therefore it is itself a unitary matrix and thus its columns are normal.

($\Leftarrow$)

Base case $d = 0$.

From (45) P and Q are just complex constants and any pair of complex constants can be parametrized as (46) and (47).

Induction step

Suppose that the theorem holds for the polynomials of degree $d - 1$ $\hat{P}$ and $\hat{Q}$. Given any target polynomials P and Q we show that there exists always a set of parameters θ_d, ϕ_d such that

$$\begin{pmatrix} \hat{P}(U) \\ \hat{Q}(U) \end{pmatrix} = A^\dagger R(\theta_d, \phi_d, 0)^\dagger \begin{pmatrix} P(U) \\ Q(U) \end{pmatrix} \quad (48)$$

By developing the latter equation we get

$$\hat{P}(U) = e^{-i\phi_d} \cos(\theta_d) U^\dagger P(U) + \sin(\theta_d) U^\dagger Q(U) \quad (49)$$

$$\hat{Q}(U) = e^{-i\phi_d} \sin(\theta_d) P(U) - \cos(\theta_d) Q(U) \quad (50)$$

For the induction to hold we need to prove the following 2 points about $\hat{P}$ and $\hat{Q}$:

- 1) Satisfy the norm condition.
- 2) Have degree at most $d - 1$.

1. Since we consider the unit circle we evaluate the polynomials on it, thus we parametrize the input as $e^{i\ell}$

$$\hat{P}(e^{i\ell}) = e^{-i\ell} \cos(\theta_\ell) e^{-i\ell} P(e^{i\ell}) + \sin(\theta_\ell) e^{-i\ell} Q(e^{i\ell}),$$

$$\hat{Q}(e^{i\ell}) = e^{-i\ell} \sin(\theta_\ell) P(e^{i\ell}) - \cos(\theta_\ell) Q(e^{i\ell}).$$

Let's apply the following parametrization

$$\begin{aligned} p &= P(e^{i\ell}), \quad q = Q(e^{i\ell}), \\ c &= \cos(\theta_\ell), \quad s = \sin(\theta_\ell), \quad \phi = \theta_\ell. \end{aligned}$$

We get

$$\hat{P} = e^{-i\ell} (e^{-i\phi} cp + sq),$$

$$\hat{Q} = e^{-i\phi} sp - cq.$$

$$\begin{aligned}
|\widehat{P}|^2 &= \widehat{P}\widehat{P}^* \\
&= [e^{-i\ell} (e^{-i\phi}cp + sq)] [e^{i\ell} (e^{i\phi}cp^* + sq^*)] \\
&= (e^{-i\phi}cp + sq) (e^{i\phi}cp^* + sq^*) \\
&= c^2|p|^2 + s^2|q|^2 + cse^{-i\phi}pq^* + cse^{i\phi}p^*q. \\
|\widehat{Q}|^2 &= \widehat{Q}\widehat{Q}^* \\
&= (e^{-i\phi}sp - cq) (e^{i\phi}sp^* - cq^*) \\
&= s^2|p|^2 + c^2|q|^2 - cse^{-i\phi}pq^* - cse^{i\phi}p^*q. \\
|\widehat{P}|^2 + |\widehat{Q}|^2 &= |p|^2(c^2 + s^2) + |q|^2(c^2 + s^2) \\
&= |p|^2 + |q|^2 \\
&= 1.
\end{aligned}$$

2. For this condition to be true we must choose θ_d and ϕ_d so that the $U^\dagger = U^{-1}$ term appearing in $\widehat{P}$ is canceled and the U^d term appearing in $\widehat{Q}$ is canceled out.

However, before searching for those angles an important consideration has to be done. We can further develop $|P(e^{i\ell})|^2 + |Q(e^{i\ell})|^2 = 1$ into

$$\left(\sum_{n=0}^d a_n e^{i\ell n}\right) \left(\sum_{n=0}^d a_n^* e^{-i\ell n}\right) + \left(\sum_{n=0}^d b_n e^{i\ell n}\right) \left(\sum_{n=0}^d b_n^* e^{-i\ell n}\right) = 1 \quad (51)$$

The term of the LHS with the largest power is

$$(a_d a_0^* + b_d b_0^*) e^{id\ell} \quad (52)$$

A necessary condition for (51) to be satisfied is that

$$a_d a_0^* = -b_d b_0^* \quad (53)$$

So that the largest power term gets canceled.

Now from (49) and (50) we get that the highest coefficient of $\widehat{Q}$ is

$$\begin{aligned}
&e^{-i\phi_d} \sin(\theta_d) a_d - \cos(\theta_d) b_d \\
&= \sin(\theta_d) e^{-i\phi_d} b_d \left(\frac{a_d}{b_d} - e^{i\phi_d} \frac{\cos(\theta_d)}{\sin(\theta_d)} \right)
\end{aligned}$$

And that the lowest coefficient of $\widehat{P}$ is

$$\begin{aligned}
& e^{-\phi_d} \cos(\theta_d) a_0 - \cos(\theta_d) b_0 \\
& = \sin(\theta_d) a_s \left(e^{-i\phi_d} \frac{\cos(\theta_d)}{\sin(\theta_d)} + \frac{b_0}{a_0} \right)
\end{aligned}$$

We want those 2 terms to be equal to 0. To do this we choose the following for the parameters:

$$\tan(\theta_d) = \frac{|b_d|}{|a_d|} \implies \theta_d = \tan^{-1}\left(\frac{|b_d|}{|a_d|}\right) \quad (54)$$

$$\phi_d = \arg\left(\frac{a_d}{b_d}\right) \quad (55)$$

Then the following conditions are satisfied.

$$(A) \quad e^{i\theta_d} \frac{\cos(\theta_d)}{\sin(\theta_d)} = \frac{a_d}{b_d}$$

$$\frac{a_d}{b_d} = \left| \frac{a_d}{b_d} \right| e^{i \arg(\frac{a_d}{b_d})}$$

Since

$$\theta_d = \tan^{-1} \left(\frac{|b_d|}{|a_d|} \right)$$

We get

$$\frac{\cos(\theta_d)}{\sin(\theta_d)} = \frac{|a_d|}{|b_d|}$$

From this and (55) condition (A) follows.

$$(B) \quad e^{-i\theta_0} \frac{\cos(\theta_0)}{\sin(\theta_0)} = -\frac{b_0}{a_0}$$

$$\text{From (53)} \quad \frac{a_d}{b_d} = -\frac{b_0^*}{a_0^*}$$

$$\text{and from (A) we get that} \quad e^{i\theta_d} \frac{\cos(\theta_d)}{\sin(\theta_d)} = -\frac{b_0^*}{a_0^*}$$

This implies (B)

□

Even though we can implement and apply a polynomial P to a unitary operator, there is still a constraint which might prevent us from doing so, i.e, the existence of Q after having chosen P . The following theorem and corollary solve this problem.

Theorem 7.2. *Given $\mathbb{T} = \{x \in \mathbb{C} : |x| = 1\}$ and $P \in \mathbb{C}[x]$ we have that*

$$\begin{aligned} \forall x \in \mathbb{T}, |P(x)|^2 \leq 1 &\implies \\ \exists Q \in \mathbb{C}[x] : \deg(P) = \deg(Q) \wedge \forall x \in \mathbb{T} |P(x)|^2 + |Q(x)|^2 &= 1 \end{aligned}$$

Corollary 7.3.

$$\begin{aligned} \forall P \in \mathbb{C}[x], \text{ with } \deg(P) = d \text{ and } |P(x)| \leq 1 \forall x \in \mathbb{T}, \\ \exists \vec{\theta} \in \mathbb{R}^{d+1}, \exists \lambda, \vec{\phi} \in \mathbb{R} : \end{aligned}$$

$$\begin{pmatrix} P(U) & \cdot \\ \cdot & \cdot \end{pmatrix} = \left(\prod_{j=1}^d R(\theta_j, \phi_j, 0) A \right) R(\theta_0, \phi_0, \lambda).$$

Where $\mathbb{T}$ is the complex unit circle, A and R are the unitary operators defined respectively in (40) and (41).

C Conclusion

Any polynomial of degree d satisfying the unit circle constraint can be applied to any unitary matrix. To actually apply the polynomial to the state vector we encode it in the following way:

$$(\langle 0| \otimes I)G(|0\rangle \otimes I) = P(U) \tag{56}$$

Proof.

$$\begin{aligned} &(\langle 0| \otimes I)G(|0\rangle \otimes I)|\psi\rangle \\ &= (\langle 0| \otimes I)G(|0\rangle \otimes |\psi\rangle) \\ &= (\langle 0| \otimes I) \begin{pmatrix} P(U) & \cdot \\ Q(U) & \cdot \end{pmatrix} \begin{pmatrix} |\psi\rangle \\ 0 \end{pmatrix} \\ &= (\langle 0| \otimes I)(|0\rangle \otimes P(U)|\psi\rangle + |1\rangle \otimes Q(U)|\psi\rangle) \\ &= P(U)|\psi\rangle \end{aligned}$$

□

8 Generalized Quantum Eigenvalue Transform (GQET)

The goal of GQET is to transform the eigenvalues of a symmetric operator encoded through a symmetric projected unitary encoding.

The following theorem formalizes how applying Generalized Quantum Signal Processing to the qubitized walk operator $\mathcal{W} = R_{\square}U$ yields a projected unitary encoding of a polynomial $v(A)$ of the symmetric operator A , effectively transforming its eigenvalues according to v .

Theorem 8.1. *Let $(U, \square)$ be a symmetric projected encoding of the symmetric matrix A . Consider a degree- d complex polynomial written in the basis of Chebyshev polynomial*

$$v(x) = \sum_{n=0}^d a_n T_n(x), \quad (57)$$

Consider its associated signal processing polynomial

$$\Upsilon(z) = \sum_{n=0}^d a_n z^n \quad (58)$$

Assume that $\max_{z \in \partial B_1(0)} |\Upsilon(z)| \leq 1$. Let $\mathcal{G}$ be the generalized quantum signal processing of $\mathcal{W} = R_{\square}U$ applying the polynomial Υ . Then, $(\mathcal{G}, |0\rangle \otimes \square)$ is a projected unitary encoding of $v(A)$, more explicitly

$$\langle 0| \otimes \square^\dagger \mathcal{G} |0\rangle \otimes \square = v(A) \quad (59)$$

Proof. Let $|v\rangle$ be eigenvector of A with eigenvalues λ .

Let $|\mu^\pm\rangle$ be eigenvectors of $\mathcal{W}$ with eigenvalues $e^{\pm i\theta}$.

We prove the equality by comparing the matrices element-wise.

$$\begin{aligned} \langle v_j | \langle 0| \otimes \square^\dagger \mathcal{G} |0\rangle \otimes \square |v_i\rangle &\stackrel{(a)}{=} \frac{\langle \mu_j^+ | - \langle \mu_j^- |}{-i\sqrt{2}} (\langle 0| \otimes 1 \mathcal{G} |0\rangle \otimes 1) \frac{|\mu_i^+\rangle - |\mu_i^-\rangle}{i\sqrt{2}} \\ &\stackrel{(b)}{=} \frac{1}{2} (\langle \mu_j^+ | - \langle \mu_j^- |) [(\Upsilon(e^{i\theta_i}) |\mu_i^+\rangle - \Upsilon(e^{-i\theta_i}) |\mu_i^-\rangle)] \\ &= \frac{\Upsilon(e^{i\theta_i}) + \Upsilon(e^{-i\theta_i})}{2} \delta_{ij} \\ &= \frac{1}{2} \sum_{n=0}^d a_n (e^{in\theta_i} + e^{-in\theta_i}) \delta_{ij} \\ &= \sum_{n=0}^d a_n \cos(n\theta_i) \delta_{ij} \\ &\stackrel{(c)}{=} \sum_{n=0}^d a_n T_n(\cos(\theta_i)) \delta_{ij} \\ &= v(\cos(\theta_i)) \delta_{ij} \\ &\stackrel{(d)}{=} v(\lambda_i) \delta_{ij} \end{aligned}$$

Where the equalities follow from:

- a Since $\mathcal{G}$ is the signal processing of $\mathcal{W}$, we want the eigenvectors of $\mathcal{W}$ to appear. We use this identity

$$\begin{aligned}\frac{|\mu^+\rangle - |\mu^-\rangle}{i\sqrt{2}} &= \frac{1}{i\sqrt{2}} \frac{1}{\sqrt{2}\sin(\theta)} (e^{i\theta} - U - e^{-i\theta} + U)\square|v\rangle \\ &= \frac{1}{i\sqrt{2}} \frac{1}{\sqrt{2}\sin(\theta)} (i2\sin(\theta))\square|v\rangle \\ &= \square|v\rangle\end{aligned}$$

- b By construction, $\langle 0| \otimes 1\mathcal{G}|0\rangle \otimes 1$ to the eigenvectors of $\mathcal{W}$ corresponds to applying the signal processing polynomial Υ to the corresponding eigenvalues.
- c We use the definition of the Chebyshev polynomials of the first kind

$$T_n(\cos(\theta)) = \cos(n\theta)$$

- d We use $\theta = \cos^{-1}(\lambda)$ from Proposition 6.2

□

A GQET of the Flat Discriminant

In this work, GQET is used for the flat discriminant. More precisely, once the flat discriminant

$$D(x, y) = \sqrt{P(x, y)P(y, x)}$$

is encoded by a SPUE $(S, \square_P)$, we identify

$$A = D.$$

The corresponding qubitized walk operator is

$$\mathcal{W} = R_{\square_P} S, \quad R_{\square_P} = 2\square_P \square_P^\dagger - I.$$

If D has eigenvalue $\lambda = \cos(\theta)$, then $\mathcal{W}$ has corresponding eigenvalues $e^{\pm i\theta}$. Therefore, applying a signal-processing polynomial Υ to $\mathcal{W}$ implements the associated Chebyshev polynomial v on the eigenvalues of D . In this way, GQET allows us to construct a projected encoding of $v(D)$.

This is useful because, by choosing v to approximate a projector onto the leading eigenspace of D , we can approximately encode the projector onto the stationary distribution $|\pi\rangle$.

9 Generalized Quantum Singular Value Transform (GQSVT)

The purpose of GQSVT is to extend polynomial transformations from eigenvalues to singular values. This is the transformation needed for non-symmetric operators, such as the curved discriminant.

A Hermitianization

GQET applies to symmetric operators, while the operator we want to transform may be rectangular or non-symmetric. Hermitianization converts such an operator A into a larger symmetric matrix $\bar{A}$ whose eigenvalues are the signed singular values of A . This allows us to apply eigenvalue transformation techniques to singular values.

We now formalize this construction with a definition.

Definition 9.1. Let A be $N_L \times N_R$ complex matrix, with corresponding PUE $(U, \square_L, \square_R)$. It is possible to construct a symmetric projected unitary encoding $(\bar{U}, \bar{\square})$ of the $\bar{N} \times \bar{N}$, with $\bar{N} = N_L + N_R$, matrix

$$\bar{A} = \begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix}. \quad (60)$$

The symmetric projected unitary has double the dimensions of U and is defined by

$$\bar{U} = (|0\rangle\langle 0| \otimes U + |1\rangle\langle 1| \otimes U^\dagger), \quad (61)$$

and the partial isometry is defined by

$$\bar{\square} = |0\rangle\langle 0| \otimes \square_L + |1\rangle\langle 1| \otimes \square_R. \quad (62)$$

The qubitized walk operator can be constructed as

$$\begin{aligned} \mathcal{W} &= R_{\bar{\square}} \bar{U} \\ R_{\bar{\square}} &= |0\rangle\langle 0| \otimes R_{\square_L} + |1\rangle\langle 1| \otimes R_{\square_R} \end{aligned}$$

The following proposition shows how the eigenvalues and eigenvectors of the Hermitianized matrix $\bar{A}$ are related to the singular values and singular vectors of the original matrix A .

Proposition 9.2. *Let a be an arbitrary complex $N_L \times N_R$ matrix. Denote the right singular vectors of A by $|v_i\rangle$ and its left singular vectors by $|w_i\rangle$. The eigen-decomposition of the hermitianization $\bar{A}$ has eigenvector-eigenvalue decomposition given by*

$$\bar{A} \frac{1}{\sqrt{2}} \begin{pmatrix} |w_i\rangle \\ \pm |v_i\rangle \end{pmatrix} = \pm \sigma_i \frac{1}{\sqrt{2}} \begin{pmatrix} |w_i\rangle \\ \pm |v_i\rangle \end{pmatrix}. \quad (63)$$

Proof. It's a straightforward computation, after we use the definition of $\bar{A}$:

$$\begin{aligned}
\bar{A} \frac{1}{\sqrt{2}} \begin{pmatrix} |w_i\rangle \\ \pm |v_i\rangle \end{pmatrix} &= \begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} |w_i\rangle \\ \pm |v_i\rangle \end{pmatrix} \\
&= \frac{1}{\sqrt{2}} \begin{pmatrix} \pm A|v_i\rangle \\ A^\dagger |w_i\rangle \end{pmatrix} \\
&= \frac{1}{\sqrt{2}} \begin{pmatrix} \pm \sigma_i |w_i\rangle \\ \sigma_i |v_i\rangle \end{pmatrix} \\
&= \pm \sigma_i \frac{1}{\sqrt{2}} \begin{pmatrix} |w_i\rangle \\ \pm |v_i\rangle \end{pmatrix}
\end{aligned}$$

□

B Matrix Polynomials

For GQSVT, the polynomial is applied to the singular values of A , not to its eigenvalues. Since A maps right singular vectors to left singular vectors, we need a definition of matrix polynomials that specifies how even and odd powers act on these two spaces.

Before formalizing this, we note that singular value transformations must distinguish between even and odd powers because they act differently on the left and right singular-vector spaces.

Let

$$A = \sum_{i=1}^r \sigma_i |w_i\rangle \langle v_i|$$

be the singular value decomposition of A , where $|v_i\rangle \in \mathbb{C}^{N_R}$ are right singular vectors and $|w_i\rangle \in \mathbb{C}^{N_L}$ are left singular vectors. Then

$$A|v_i\rangle = \sigma_i |w_i\rangle, \quad A^\dagger |w_i\rangle = \sigma_i |v_i\rangle.$$

Hence

$$A^\dagger A|v_i\rangle = \sigma_i^2 |v_i\rangle, \quad AA^\dagger |w_i\rangle = \sigma_i^2 |w_i\rangle.$$

Thus even powers preserve the corresponding singular-vector space, whereas odd powers map between the right and left singular-vector spaces. This motivates the following precise definition.

Definition 9.3. Let A be an arbitrary complex $N_L \times N_R$ matrix. It admits singular value decomposition $A = W^\dagger D V$, with $A|v_i\rangle = \sigma_i |w_i\rangle$. Here, V is a $N_R \times N_R$ unitary matrix, W is a $N_L \times N_L$ unitary and D a $N_L \times N_R$ with the $\min(N_L, N_R)$ singular values $(\sigma_i)_{i=0}^{\min(N_L, N_R)-1}$ arranged on the diagonal. The singular value transformations by even $p_e(x)$ or odd $p_o(x)$ polynomials are defined by:

$$\begin{cases} p_o(A) = W^\dagger p_o(D) V \\ p_e(A) = V^\dagger p_e(D') V \end{cases} \quad (64)$$

with D' being D restricted to a $N_R \times N_R$ diagonal matrix which keeps all the singular values $(\sigma_i)_{i=0}^{\min(N_L, N_R)-1}$.

C GQSVT

After hermitianization, applying GQET to $\bar{A}$ induces a singular-value transformation of A . The next proposition makes explicit how the even and odd parts of the polynomial act on the left and right singular-vector spaces.

Proposition 9.4. *Consider a polynomial $p(x)$ having for even part $p_e(x)$ and odd part $p_o(x)$. Let $\mathcal{G}$ be the generalized quantum eigenvalue transform applying $p(x)$ to the hermitianized projected unitary encoding $\bar{U}$ of $\bar{A}$. Then,*

$$\langle 0| \otimes \bar{\square}^\dagger \mathcal{G} |0\rangle \otimes \bar{\square} = \begin{pmatrix} W^\dagger p_e(D)W & p_o(A) \\ (p_o(A))^\dagger & p_e(A) \end{pmatrix}. \quad (65)$$

Proof. We prove the proposition by considering the action of $(\langle 0| \otimes \bar{\square}^\dagger) \mathcal{G} (|0\rangle \otimes \bar{\square})$ on basis vectors of the form

$$\begin{pmatrix} |w_i\rangle \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ |v_i\rangle \end{pmatrix}. \quad (66)$$

We start by computing the effect of the first basis vector on the RHS,

$$\begin{aligned} \langle 0| \otimes \bar{\square}^\dagger \mathcal{G} |0\rangle \otimes \bar{\square} \begin{pmatrix} |w_i\rangle \\ 0 \end{pmatrix} &\stackrel{(a)}{=} \frac{1}{2} \langle 0| \otimes \bar{\square}^\dagger \mathcal{G} |0\rangle \otimes \bar{\square} \left[\begin{pmatrix} |w_i\rangle \\ |v_i\rangle \end{pmatrix} + \begin{pmatrix} |w_i\rangle \\ -|v_i\rangle \end{pmatrix} \right] \\ &= \frac{1}{2} \left[p(\sigma_i) \begin{pmatrix} |w_i\rangle \\ |v_i\rangle \end{pmatrix} + p(-\sigma_i) \begin{pmatrix} |w_i\rangle \\ -|v_i\rangle \end{pmatrix} \right] \\ &\stackrel{(b)}{=} \frac{1}{2} \left[(p_e(\sigma_i) + p_o(\sigma_i)) \begin{pmatrix} |w_i\rangle \\ |v_i\rangle \end{pmatrix} + (p_e(\sigma_i) - p_o(\sigma_i)) \begin{pmatrix} |w_i\rangle \\ -|v_i\rangle \end{pmatrix} \right] \\ &= \begin{pmatrix} p_e(\sigma_i) |w_i\rangle \\ p_o(\sigma_i) |v_i\rangle \end{pmatrix} \end{aligned}$$

Where the equalities come from:

- a We rewrite the basis vectors in terms of the eigenvectors of $\bar{A}$, so that we can simply apply the polynomial to their corresponding eigenvalues.
- b We decompose the polynomial into its even and odd component, and we use the properties of odd and even functions:

$$p_e(x) = p_e(-x), \quad p_o(x) = -p_o(-x) \quad (67)$$

We compute the effect of the first basis vector on the LHS, and conclude it is equal to its effect on the RHS.

$$\begin{pmatrix} W^\dagger p_e(D)W & p_o(A) \\ (p_o(A))^\dagger & p_e(A) \end{pmatrix} \begin{pmatrix} |w_i\rangle \\ 0 \end{pmatrix} = \begin{pmatrix} W^\dagger p_e(D)W |w_i\rangle \\ (p_o(A))^\dagger |w_i\rangle \end{pmatrix} \stackrel{(a)}{=} \begin{pmatrix} p_e(\sigma_i) |w_i\rangle \\ (p_o(A))^\dagger |w_i\rangle \end{pmatrix} \stackrel{(b)}{=} \begin{pmatrix} p_e(\sigma_i) |w_i\rangle \\ p_o(\sigma_i) |v_i\rangle \end{pmatrix}$$

Where the equalities come from:

a We expand the diagonal matrix D to apply the polynomial to its diagonal elements.

$$\begin{aligned}
W^\dagger p_e(D)W|w_i\rangle &= \sum_j p_e(\sigma_j)W^\dagger|e_j\rangle\langle e_j|W|w_i\rangle \\
&= \sum_j p_e(\sigma_j)W^\dagger|e_j\rangle\langle e_j|e_i\rangle \\
&= p_e(\sigma_i)W^\dagger|e_i\rangle \\
&= p_e(\sigma_i)|w_i\rangle
\end{aligned}$$

b We apply the odd polynomial to A using Definition 9.3

$$\begin{aligned}
(p_o(A))^\dagger|w_i\rangle &= (W^\dagger p_o(D)V)^\dagger|w_i\rangle \\
&= V^\dagger p_o(D)^\dagger W|w_i\rangle \\
&= \sum_j p_o(\sigma_j)V^\dagger|e_j\rangle\langle e_j|W|w_i\rangle \\
&= \sum_j p_o(\sigma_j)|v_j\rangle\langle e_j|e_i\rangle \\
&= p_o(\sigma_i)|v_i\rangle
\end{aligned}$$

Then, we compute the effect of the second basis vector on the RHS,

$$\begin{aligned}
\langle 0|\otimes\bar{\square}^\dagger\mathcal{G}|0\rangle\otimes\bar{\square}\begin{pmatrix} 0 \\ |v_i\rangle \end{pmatrix} &= \frac{1}{2}\langle 0|\otimes\bar{\square}^\dagger\mathcal{G}|0\rangle\otimes\bar{\square}\left[\begin{pmatrix} |w_i\rangle \\ |v_i\rangle \end{pmatrix}-\begin{pmatrix} |w_i\rangle \\ -|v_i\rangle \end{pmatrix}\right] \\
&= \frac{1}{2}\left[p(\sigma_i)\begin{pmatrix} |w_i\rangle \\ |v_i\rangle \end{pmatrix}-p(-\sigma_i)\begin{pmatrix} |w_i\rangle \\ -|v_i\rangle \end{pmatrix}\right] \\
&= \frac{1}{2}\left[(p_e(\sigma_i)+p_o(\sigma_i))\begin{pmatrix} |w_i\rangle \\ |v_i\rangle \end{pmatrix}-(p_e(\sigma_i)-p_o(\sigma_i))\begin{pmatrix} |w_i\rangle \\ -|v_i\rangle \end{pmatrix}\right] \\
&= \begin{pmatrix} p_o(\sigma_i)|w_i\rangle \\ p_e(\sigma_i)|v_i\rangle \end{pmatrix}
\end{aligned}$$

We compute the effect of the second basis vector on the LHS, and conclude it is equal to its effect on the RHS.

$$\begin{pmatrix} W^\dagger p_e(D)W & p_o(A) \\ (p_o(A))^\dagger & p_e(A) \end{pmatrix} \begin{pmatrix} 0 \\ |v_i\rangle \end{pmatrix} = \begin{pmatrix} p_o(A)|v_i\rangle \\ p_e(A)|v_i\rangle \end{pmatrix} \stackrel{(a)}{=} \begin{pmatrix} p_o(\sigma_i)|w_i\rangle \\ p_e(A)|v_i\rangle \end{pmatrix} \stackrel{(b)}{=} \begin{pmatrix} p_o(\sigma_i)|w_i\rangle \\ p_e(\sigma_i)|v_i\rangle \end{pmatrix}$$

Where the equalities come from:

a We apply the odd polynomial to A using Definition 9.3

$$\begin{aligned}
p_o(A)|v_i\rangle &= W^\dagger p_o(D)V|v_i\rangle \\
&= \sum_j p_o(\sigma_j)W^\dagger|e_j\rangle\langle e_j|V|v_i\rangle \\
&= \sum_j p_o(\sigma_j)|w_j\rangle\langle e_j|e_i\rangle \\
&= p_o(\sigma_i)|w_i\rangle
\end{aligned}$$

b We apply the even polynomial to A using Definition 9.3

$$\begin{aligned}
p_e(A)|v_i\rangle &= V^\dagger p_e(D')V|v_i\rangle \\
&= \sum_j p_e(\sigma_j) V^\dagger |e_j\rangle \langle e_j| V |v_i\rangle \\
&= \sum_j p_e(\sigma_j) |v_j\rangle \langle e_j| e_i\rangle \\
&= p_e(\sigma_i) |v_i\rangle
\end{aligned}$$

□

10 GQSVT of the Curved Discriminant

In this setting, GQSVT is used for the curved discriminant. More precisely, once the curved discriminant

$$\mathcal{D}(x, y) = \sqrt{P(x, y)P^*(y, x)}$$

is encoded by a PUE $(S, \square_P, \square_{P^*})$, we identify

$$A = \mathcal{D}.$$

Since $\mathcal{D}$ is in general not symmetric, we cannot directly apply GQET. Instead, we apply GQSVT, which acts on the singular values of $\mathcal{D}$.

In the hermitianized construction, an additional ancilla qubit distinguishes the blocks appearing in (65). By projecting this ancilla onto the state $|1\rangle$, we select the block containing the even polynomial acting on the right singular-vector space. Thus,

$$(\langle 01| \otimes \bar{\square}^\dagger) \mathcal{G}(|01\rangle \otimes \bar{\square}) = (\langle 01| \otimes \square_P^\dagger) \mathcal{G}(|01\rangle \otimes \square_P) = V^\dagger p_e(\Sigma) V = p_e(\mathcal{D}),$$

where $\mathcal{D} = W^\dagger \Sigma V$ is the singular value decomposition of $\mathcal{D}$. By projecting the second ancilla qubit onto $|1\rangle$, we effectively select only the right block $\square_P$ of the Hermitianized partial isometry $\bar{\square}$, which corresponds to the forward chain.

Thus, in this construction, only even polynomials give a direct encoding of the desired singular-value transformation. We choose p_e to be the even fast-forwarding polynomial. Since its scaling factor is equal to 1, no additional normalization loss appears. For the curved discriminant, the singular value 1 corresponds to the stationary coherent state $|\pi\rangle$. Therefore, applying GQSVT with the fast-forwarding polynomial gives an approximate projected unitary encoding of $|\pi\rangle\langle\pi|$.

11 Sampling from the distribution

We have seen from GQET that

$$(\langle 0| \otimes \square_p^\dagger) \mathcal{G}(|0\rangle \otimes \square_p) = v(D) = |\pi\rangle \langle \pi| \quad (68)$$

And from GQSVT that

$$\left(\langle 01| \otimes \square_P^\dagger \right) \mathcal{G}(|01\rangle \otimes \square_P) = v_e(\mathcal{D}) = |\pi\rangle \langle \pi|$$

The question now is how to sample from the stationary distribution π . The answer is straightforward: we prepare an initial state that has nonzero overlap with $|\pi\rangle$ and feed it into the circuit. While this is not the only possible choice, we select the equally weighted superposition state, obtained by applying a Hadamard gate to all qubits initialized in $|0\rangle$, since this guarantees a nonzero overlap with the target stationary state. So we get:

$$|\text{initial}\rangle = H^{\otimes n} |0\rangle^{\otimes n} = \frac{1}{2^{\frac{n}{2}}} \sum_{x \in \{0,1\}^n} |x\rangle$$

and

By feeding this state to our circuit, for the second method (GQET) we get:

$$(\langle 0| \otimes \square_P^\dagger) \mathcal{G}(|0\rangle \otimes \square_P) |\text{initial}\rangle = |\pi\rangle \langle \pi | \text{initial}\rangle = |\pi\rangle, \quad (69)$$

and for the first method (GQSVT) we get:

$$(\langle 01| \otimes \square_P^\dagger) \mathcal{G}(|01\rangle \otimes \square_P) |\text{initial}\rangle = |\pi\rangle \langle \pi | \text{initial}\rangle = |\pi\rangle, \quad (70)$$

Where the last equality follows from the fundamental fact that all quantum states are normalized.

12 Experiment

In this section, we demonstrate the implementation of the algorithm for both reversible and nonreversible Markov chains using the Python library Qiskit. Since the reversible case is simpler to implement, we begin with it before moving on to the nonreversible case.

A Implementation for reversible Markov chain

Since the Markov chain is reversible, its flat discriminant D has a leading eigenvalue $\lambda_1 = 1$, with the corresponding eigenvector given by the stationary distribution $|\pi\rangle$. Moreover, D admits a symmetric projected unitary encoding (SPUE) $(S, \square_P)$. Consequently, the projector onto the stationary distribution, $|\pi\rangle\langle\pi|$, can be implemented directly using GQET applied to this SPUE.

Setup

Consider the Markov chain defined by the transition matrix:

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 \\ 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

The associated stationary distribution for this chain is given by the row vector:

$$\pi = \left[\frac{1}{6} \quad \frac{1}{3} \quad \frac{1}{3} \quad \frac{1}{6} \right]$$

with the corresponding quantum state:

$$|\pi\rangle = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}$$

Our objective is to implement the following projector:

$$|\pi\rangle\langle\pi|$$

To achieve this, we construct the block-encoded framework:

$$(\langle 0| \otimes \square_P^\dagger) G(|0\rangle \otimes \square_P) = \nu(D) \tag{71}$$

using Generalized Quantum Signal Processing (GQSP). The polynomial ν is chosen such that it maps the leading eigenvalue $\lambda_1 = 1$ to 1, while all other eigenvalues are suppressed to values $|\nu(\lambda_i)| \leq \epsilon$. Therefore, the resulting operator approximates the projector onto the stationary state:

$$\nu(D) = \sum_i \nu(\lambda_i) |\lambda_i\rangle \langle \lambda_i| \approx |\lambda_1\rangle \langle \lambda_1| = |\pi\rangle \langle \pi|. \tag{72}$$

Here, we have used the fact that D is symmetric, so it admits a spectral decomposition, and that its leading eigenvector corresponds to the stationary distribution of the chain, $|\pi\rangle$.

Since our matrix is 4×4 , we can encode each state of the chain using 2 qubits, i.e., the Hilbert space for a single register is $S = \mathbb{C}^2 \otimes \mathbb{C}^2$.

Main operators definition

To build the framework, we define the core operators, starting from the partial isometry:

$$\square_P = \sum_{x \in S} \sum_{y \in S} \sqrt{P(x, y)} |x\rangle |y\rangle \langle x|$$

The swap operator:

$$S = \sum_{x, y \in S} |x, y\rangle \langle y, x|$$

By construction, this satisfies $\square_P^\dagger S \square_P = D$, where D is the flat discriminant. From these blocks, we define the reflection operator R and the qubitized walk operator $\mathcal{W}$, which must be unitary to be used in GQSP:

$$\begin{aligned} R &= 2\square_P \square_P^\dagger - I \\ W &= RS = (2\square_P \square_P^\dagger - I)S \end{aligned}$$

To apply the partial isometry $\square_P$ in a quantum circuit, it must be unitary. Since the partial isometry is not naturally unitary, we apply a unitary dilation to make it unitary, acting on the dilated space $\mathbb{C}^2 \otimes \mathbb{C}^2$. Using this dilated isometry, which we might call V , the full qubitized walk operator is constructed as the product of a reflection and a swap operation:

$$W = (2VV^\dagger - I)S$$

Chebyshev polynomial

Our goal is to filter the spectrum of D to isolate the leading eigenvalue 1, which corresponds to the stationary distribution, while suppressing all other eigenvalues. Let $\delta = \sigma_1 - \sigma_2$ denote the spectral gap of D . In our example, $\delta = 1 - 0.75 = 0.25$.

We choose an error bound $\epsilon = 0.125$ to quantify how much residual amplitude from the non-leading eigenvalues is tolerated after applying the polynomial filter.

The minimum degree d of the Chebyshev polynomial required to achieve this suppression is given by

$$d = \left\lceil \frac{\text{arccosh}(1/\epsilon)}{\text{arccosh}(1/(1 - \Delta))} \right\rceil = 4.$$

We define the scaling factor

$$\alpha = \cosh \left(\frac{\text{arccosh}(1/\epsilon)}{d} \right),$$

and apply the final scaled Chebyshev polynomial of the first kind to the walk operator:

$$v(x) = \epsilon T_d(\alpha x).$$

By construction, this guarantees that $v(1) = 1$ for the leading eigenvalue and that $|\nu(\lambda)| \leq \epsilon$ for all other eigenvalues λ , effectively approximating the projector onto the stationary distribution. The polynomial v is the one which will be implemented through GQSP.

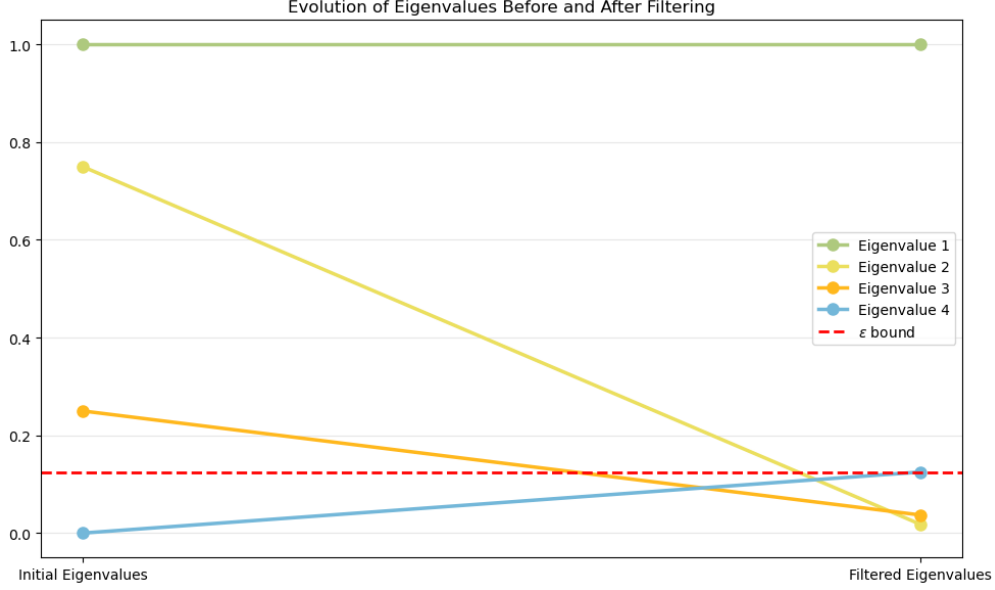


Figure 1: Evolution of eigenvalues before and after filtering. The non-leading eigenvalues are successfully bounded by the $\epsilon = 0.125$ threshold.

To implement the target quantum algorithm for a 4-state Markov chain, we allocated a total of six qubits as shown in Figure (2a):

- two qubits for the current state register (x),
- two qubits for the next state register (y),
- one ancillary qubit (s) for applying the GQSP rotations.

Next, we initialize the state to an equal superposition, ensuring that it has some overlap with $|\pi\rangle$. This is achieved by applying Hadamard gates to the qubits x_0 and x_1 , as illustrated in Figure 2b.

Once the state is correctly initialized, we apply the first transformation V to both the starting state x and the target state y according to (71), as shown in Figure 2c.

We then implement the gate sequence that applies the polynomial $v(W)$ using the Generalized Quantum Signal Processing (GQSP) framework. The rotation angles $\vec{\theta}$, $\vec{\phi}$, the phase λ , and the complementary polynomial Q with Chebyshev coefficients $\vec{q}$ are computed using the algorithms provided in [2]. The resulting values are:

$$\lambda = 0.4260285526844599$$

$$\vec{\theta} = \begin{bmatrix} 0.7421365375656978 \\ 0.7225906010558145 \\ 0.003883313351271157 \\ 0.7229092672503551 \\ 0.7460096424917101 \end{bmatrix}$$

$$\vec{\phi} = \begin{bmatrix} -1.572112848606204 \\ -1.5930629405866346 \\ -1.5459133145175012 \\ 1.5694874596723283 \\ 2.7155730913535514 \end{bmatrix}$$

$$\vec{q} = \begin{bmatrix} 0.2542 + 0.1154i \\ -0.2036 + 0.4503i \\ -0.0025 + 0.0002i \\ 0.2064 - 0.4532i \\ -0.2562 - 0.1163i \end{bmatrix}$$

Finally, we apply $V^\dagger$ and perform the measurements, still according to (71). It is important to note that the ancillary qubit must collapse to the state $|0\rangle$; if it does not, the circuit must be rerun. The complete circuit is shown in Figure 3.

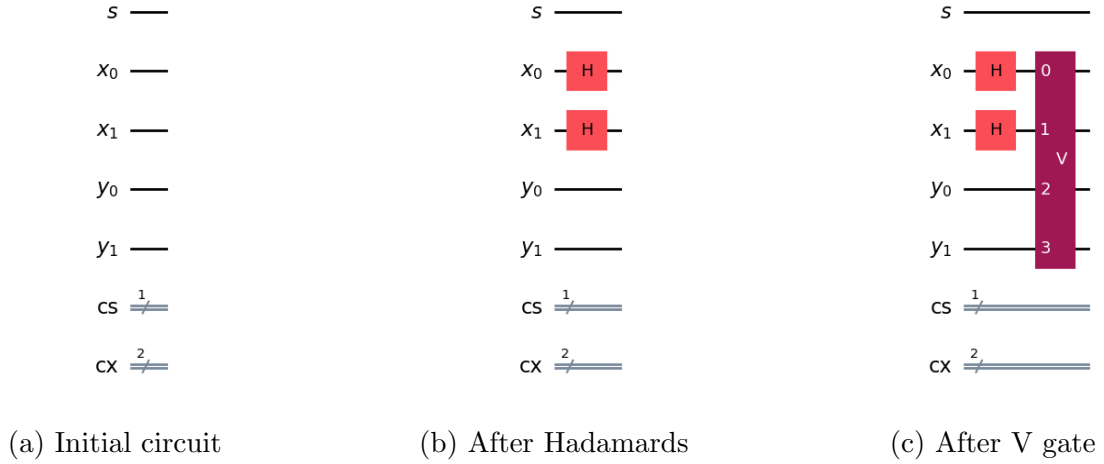


Figure 2: Circuit construction steps

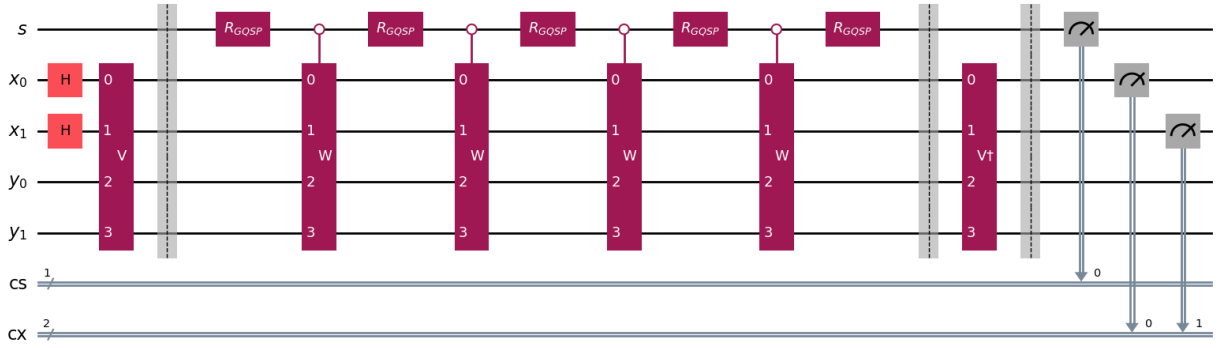


Figure 3: Final circuit, with barriers to highlight the different parts of it.

B Results - Noiseless Backend

We ran the circuit 10,000 times using the `qiskit_aer` simulator and obtained the following results:

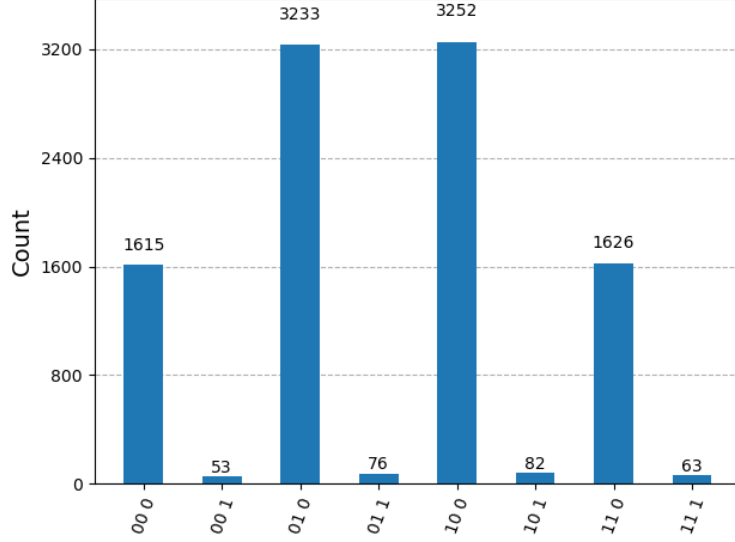


Figure 4: Measurement results obtained from running the circuit 10,000 times.

In the figure, the first two bits correspond to the measurement of the x register, while the last bit corresponds to the measurement of the ancillary qubit. As mentioned earlier, if the ancillary qubit is measured in the $|1\rangle$ state, the result is discarded. This yields a success rate for this run of the algorithm of 97.26%. The resulting probability distribution of the quantum circuit is:

$$\tilde{\pi} = [0.1660 \quad 0.3324 \quad 0.3344 \quad 0.1672] .$$

Comparing this sampled outcome to the true analytical stationary distribution:

$$\pi = \left[\frac{1}{6} \quad \frac{1}{3} \quad \frac{1}{3} \quad \frac{1}{6} \right] \approx [0.1667 \quad 0.3333 \quad 0.3333 \quad 0.1667] ,$$

we observe an exceptionally high convergence accuracy. To be more precise, the calculated Total Variation Distance d_{TV} between the target distribution and the simulated output is:

$$d_{TV} = \frac{1}{2} \sum_i |\pi_i - \tilde{\pi}_i| \approx 0.00154.$$

Thus, a quantum computer can effectively be used (assuming fault-tolerance) to sample from the stationary distribution of a reversible Markov chain with a quadratic improvement relative to the degree of the polynomial.

C Reality Check - Noisy Backends

In the previous implementation, the circuit was run on a noiseless quantum simulator, and it is evident that it successfully approximates the final state of the circuit to the coherent state of the stationary distribution $|\pi\rangle$.

However, when considering the major challenge in quantum computing -noise- the results are significantly less ideal. We ran the circuit using both a noisy simulator and a real quantum backend. Specifically, we used:

- A noisy simulator based on Aer:

```
fake_backend = GenericBackendV2(num_qubits=5)
simulator_noise = AerSimulator.from_backend(fake_backend)
```

- A real IBM backend: `ibm.kingston`.

In both cases, the final measured distribution deviates noticeably from the target stationary distribution. The results obtained with the noisy Aer simulator are shown in Figure 5, with a corresponding total variation distance $d_{TV} = 0.1436$. The results obtained using the real IBM backend are shown in Figure 6, with $d_{TV} = 0.1742$. In both histograms, the resulting distribution is approximately uniform across all states. This occurs because gate errors, decoherence, and readout noise cause the quantum state to lose the amplitude information that encodes the stationary distribution $|\pi\rangle$, effectively randomizing the outcome and producing nearly equal probabilities for all basis states.

These observations highlight the critical impact of noise on the fidelity of quantum sampling, emphasizing the importance of error mitigation and fault-tolerant techniques for accurate implementation on current quantum hardware.

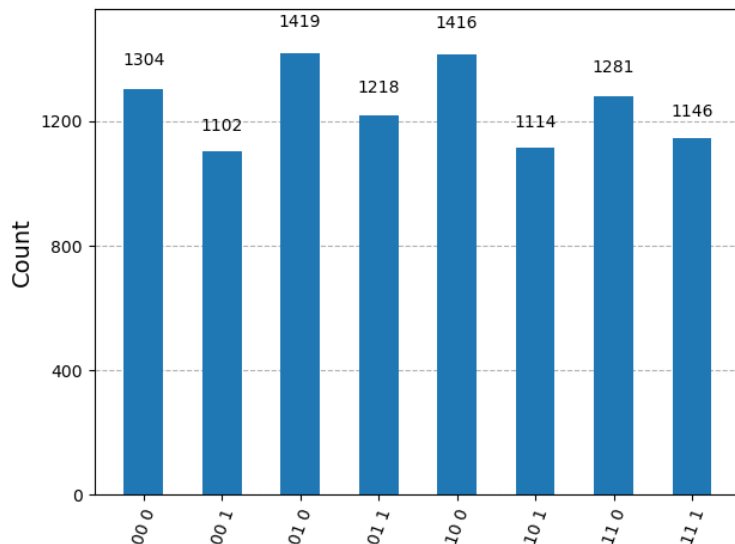


Figure 5: Measurement results on a noise simulator based on Aer.

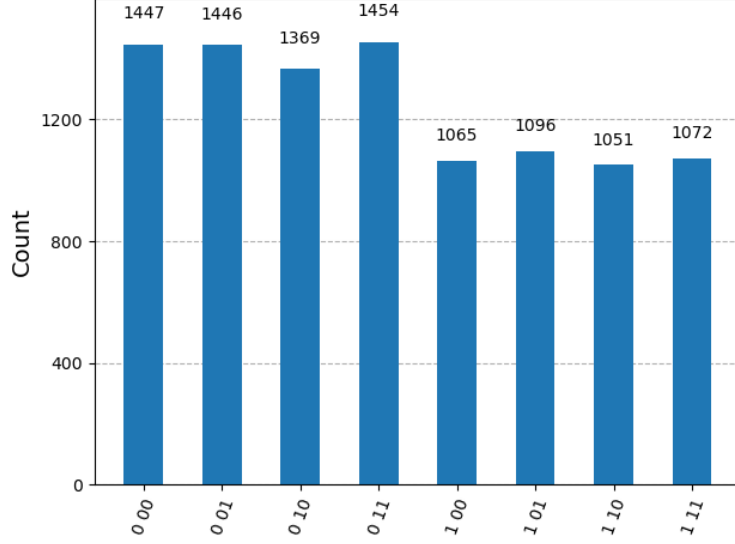


Figure 6: Measurement results on a real IBM backend.

These observations highlight the critical impact of noise on the fidelity of quantum sampling, emphasizing the importance of error mitigation and fault-tolerant techniques for accurate implementation on current quantum hardware.

D Controlled Noise Analysis

From the previous section, it is evident that the current noise levels of realistic quantum backends completely destroy the amplitude information, rendering the proposed algorithm unsuitable for practical use. Nevertheless, we aim to study the effect of controlled noise on the final distribution generated by the circuit. To this end, we focus on depolarizing errors affecting both single-qubit and two-qubit gates, vary the error probability p , and analyze the resulting total variation distance between the obtained distribution and the target stationary distribution.

From Figure 7, we observe that the total variation distance increases with the error probability. In particular, when the depolarizing error probability exceeds approximately $p \sim 0.001$, the total variation distance surpasses 0.125, which corresponds to the ε threshold used in the Chebyshev fast-forwarding polynomial. This indicates that, beyond this error rate, the noise overwhelms the spectral filtering and the algorithm no longer reliably prepares a state close to the target stationary distribution.

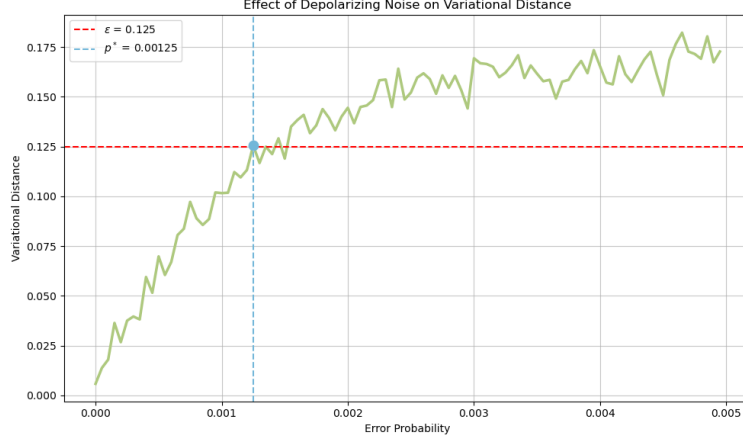


Figure 7: Total variational distance between the final distribution and the stationary distribution as a function of the error probability.

E Implementation for a non-reversible Markov chain

The previous implementation served as a toy model to test the algorithm, focusing solely on a reversible Markov chain. Since the main contribution of [1] is to extend the quantum speedup to nonreversible chains, we now focus on this more relevant case. This is achieved by employing the GQSVT instead of GQET, as we now work with the curved discriminant $\mathcal{D}$.

Setup

For the non-reversible case, we choose the following 8×8 Markov kernel:

$$P = \begin{bmatrix} 0.54989 & 0.34993 & 0.07998 & 0.01999 & 0.00005 & 0.00006 & 0.00006 & 0.00004 \\ 0.04999 & 0.59988 & 0.29994 & 0.04999 & 0.00005 & 0.00006 & 0.00006 & 0.00004 \\ 0.01999 & 0.07998 & 0.54989 & 0.34993 & 0.00005 & 0.00006 & 0.00006 & 0.00004 \\ 0.29994 & 0.04999 & 0.04999 & 0.59988 & 0.00005 & 0.00006 & 0.00006 & 0.00004 \\ 0.00017 & 0.00022 & 0.00019 & 0.00021 & 0.49960 & 0.09992 & 0.34972 & 0.04996 \\ 0.00017 & 0.00022 & 0.00019 & 0.00021 & 0.29976 & 0.49960 & 0.04996 & 0.14988 \\ 0.00017 & 0.00022 & 0.00019 & 0.00021 & 0.04996 & 0.34972 & 0.49960 & 0.09992 \\ 0.00017 & 0.00022 & 0.00019 & 0.00021 & 0.00999 & 0.04996 & 0.29976 & 0.54956 \end{bmatrix}$$

The unique stationary distribution corresponding to this kernel is:

$$\pi = [0.173115 \quad 0.217577 \quad 0.199176 \quad 0.210132 \quad 0.046778 \quad 0.055481 \quad 0.060586 \quad 0.037155]$$

This gives the normalized target quantum state:

$$|\pi\rangle = \begin{bmatrix} 0.416071 \\ 0.466452 \\ 0.446291 \\ 0.458401 \\ 0.216283 \\ 0.235544 \\ 0.246142 \\ 0.192756 \end{bmatrix}$$

Our objective is to implement the projector

$$|\pi\rangle\langle\pi|.$$

To achieve this, we construct the block-encoded framework using the Hermitianized partial isometry:

$$\left(\langle 01| \otimes \square_P^\dagger\right) \mathcal{G}(|01\rangle \otimes \square_P). \quad (73)$$

Main Operators

To implement the quantum algorithm, we first define the fundamental isometries and routing operators. We introduce the state-preparation partial isometries for both the forward transition dynamics $\square_P$ and the time-reversed dynamics $\square_{P^*}$, along with the standard register swap operator S :

$$\square_P = \sum_{x \in \mathbb{S}} |x\rangle |P(x, \cdot)\rangle \langle x|, \quad \square_{P^*} = \sum_{x \in \mathbb{S}} |x\rangle |P^*(x, \cdot)\rangle \langle x|, \quad S = \sum_{x, y \in \mathbb{S}} |x, y\rangle \langle y, x|.$$

The final filtering operator $\mathcal{G}$ is constructed using the Generalized Quantum Signal Processing (GQSP) framework. Unlike the reversible case, the Chebyshev polynomial cannot be applied directly to a flat symmetric walk. Instead, the polynomial acts on a modified, Hermitianized walk operator defined on an enlarged Hilbert space:

$$\mathcal{W} = (2\bar{\square}\bar{\square}^\dagger - I)\bar{S},$$

where the Hermitianized partial isometry $\bar{\square}$ and swap operator $\bar{S}$ are defined by introducing an additional ancilla qubit:

$$\bar{\square} = (|0\rangle\langle 0| \otimes \square_{P^*}) + (|1\rangle\langle 1| \otimes \square_P), \quad \bar{S} = \left(|0\rangle\langle 0| \otimes S + |1\rangle\langle 1| \otimes S^\dagger\right)(X \otimes I),$$

with X denoting the standard Pauli-X gate. Projecting the second ancilla qubit onto $|1\rangle$ ensures that only the right partial isometry $\square_P$ is selected, enabling the polynomial transformation to act on the curved discriminant $\mathcal{D}$.

Reversibilization time

As discussed in Section (1), before determining the degree of our polynomial, we consider the iterated kernel $P^{t_{\text{rev}}}$ instead of P . The reversibilization time t_{rev} is defined as

$$t_{\text{rev}} = \operatorname{argmax}_{k \geq 1} \frac{\gamma(P^k(P^*)^k)}{k},$$

where $\gamma(\cdot)$ denotes the spectral gap of a reversible kernel. To find t_{rev} , we performed a search up to $k_{\text{max}} = 1000$, yielding $t_{\text{rev}} = 1$.

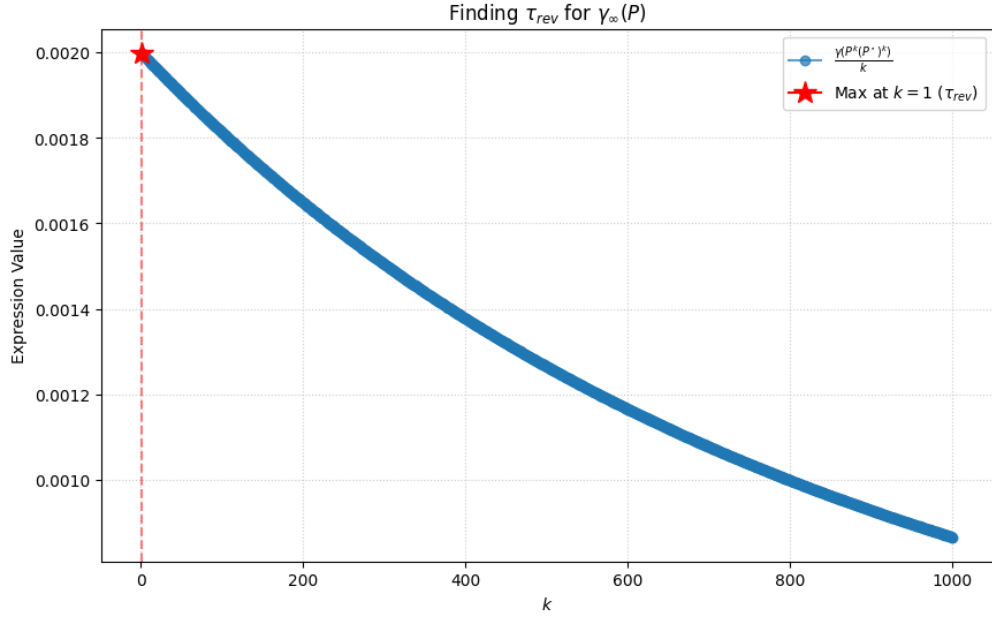


Figure 8: Evolution of the normalized spectral gap $\frac{\gamma(P^k(P^*)^k)}{k}$ as a function of k .

Chebyshev polynomial

We define the polynomial filter $\nu(x)$ that is applied via the Generalized Quantum Signal Processing framework to the Hermitized walk operator $\mathcal{W}$. To effectively isolate the stationary state while tightly bounding the remaining spectrum, we set our target error threshold to $\epsilon = 0.001$.

To achieve this desired error bound, the minimum required degree d of the Chebyshev polynomial is determined by the analytical bound:

$$d = \left\lceil \frac{\operatorname{arccosh}\left(\frac{1}{\epsilon}\right)}{\operatorname{arccosh}\left(\frac{1}{1-\Delta_{nr}}\right)} \right\rceil \approx \lceil 169.891 \rceil = 170$$

Given the calculated degree $d = 170$, we structure our filter polynomial as a scaled Chebyshev polynomial of the first kind:

$$\nu(x) = \epsilon T_d(\alpha x)$$

To compute the scaling factor α , we enforce the strict constraint that the leading singular value must map to unity ($\nu(1) = 1$), ensuring that the target stationary state is preserved with maximum probability. This design requirement yields the following relation for α :

$$\alpha = \cosh \left(\frac{\operatorname{arccosh} \left(\frac{1}{\epsilon} \right)}{d} \right)$$

By satisfying this condition, the filter guarantees that $\nu(1) = 1$ while simultaneously ensuring that all non leading eigenvalues λ are mapped as:

$$|\nu(\lambda)| \leq \epsilon = 0.001$$

Due to the very high degree ($d = 170$) required to resolve the small gap between the leading eigenvalue and the subleading ones, we omit the explicit list of coefficients; nevertheless, the rapid oscillations of this high-degree Chebyshev polynomial ensure a sharp spectral filtering that effectively isolates the leading eigenvalue, even for the non-reversible chain.

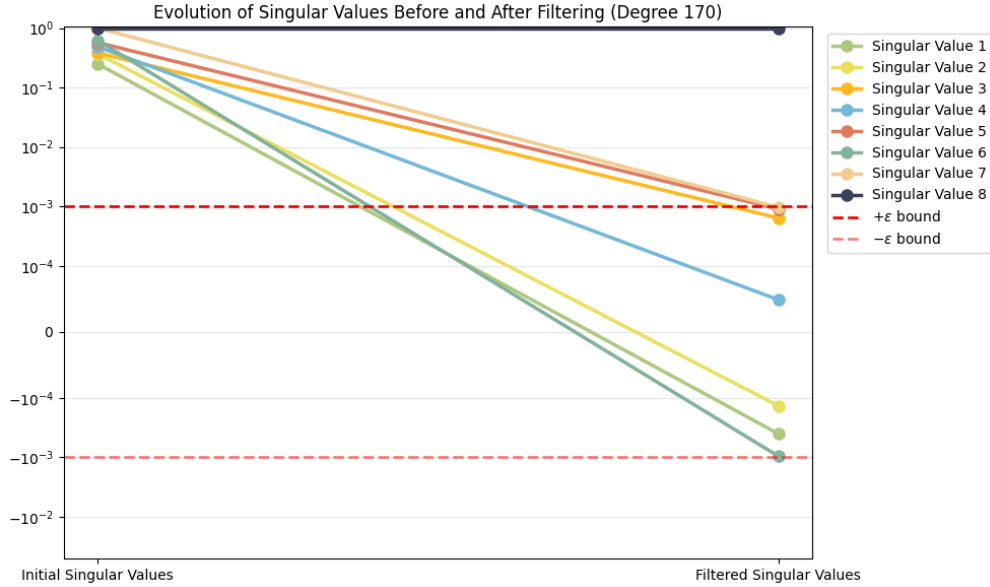


Figure 9: Spectral filtering using the $d = 170$ scaled Chebyshev polynomial, bounding sub-leading singular values within the $\epsilon = 0.01$ threshold.

Building the circuit

To physically implement the filtering polynomial for the 8-state non-reversible Markov chain, we allocate a total of 8 qubits. This increases the circuit complexity compared to the reversible case, due both to the larger state space and the additional qubit required for Hermitization of the operator.

The quantum register is partitioned as follows:

- one signal qubit (s): the target qubit upon which the parameterized (GQSP) rotations are actively applied.

- three state qubits (x): the register encoding the current position/state of the Markov chain ($2^3 = 8$ states).
- three target qubits (y): the register encoding the transition probabilities for the next step of the walk.
- one hermitization qubit (h): an additional ancillary qubit introduced strictly for the non-reversible case.

At first, we initialize our state qubits to the equally superpositioned state and the hermitization qubit to the state 1 in accordance with (73).

Then we apply the dilated unitary version of $\square_P$, indicated as V .

Next, using the algorithm provided in the GQSP paper we apply the GQSP block.

Lastly we apply $V^\dagger$ and measure the registers. The final circuit is shown in Figure 10

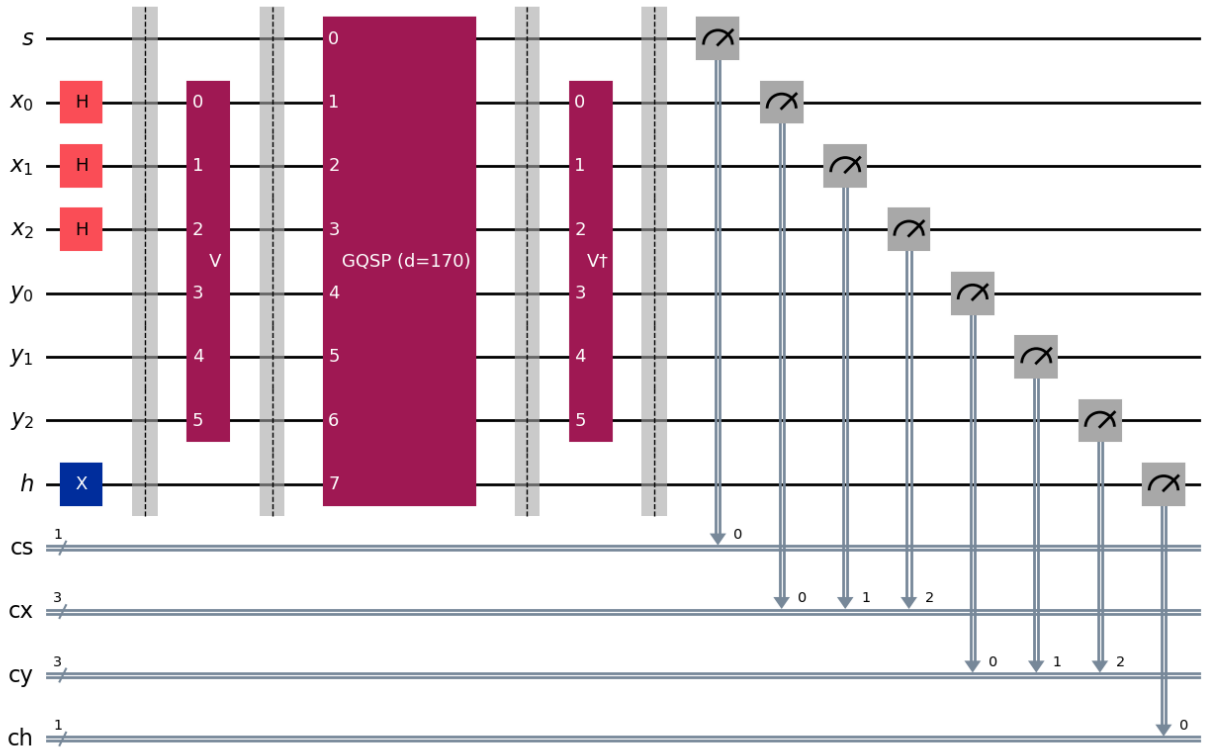


Figure 10: Final circuit for the non reversible Markov kernel

Results

To extract the filtered stationary distribution from the final quantum state, we must apply a post-selection procedure that is significantly stricter than the one utilized for the reversible chain. Because the non-reversible algorithm operates within a highly dilated Hilbert space, valid measurement outcomes require satisfying three distinct conditions simultaneously. First, the GQSP signal qubit must be measured in the $|0\rangle$ state, indicating successful

application of the filtering polynomial. Second, the Hermitization control qubit must be measured as $|1\rangle$, which projects the system onto the correct operational block. Finally, the target register must collapse entirely to the all-zero state $|000\rangle$, ensuring the quantum state is properly projected back onto the input subspace of the partial isometry. Consequently, only measurement bitstrings matching the strict configuration $|h, y, x, s\rangle = |1, 000, x, 0\rangle$ are retained, and the relative frequencies of the remaining x -register values yield the empirical probability distribution $\tilde{\pi}$.

In the same way as for the reversible case we run the circuit 10 000 times using `qiskit_aer` simulator and got the following results:

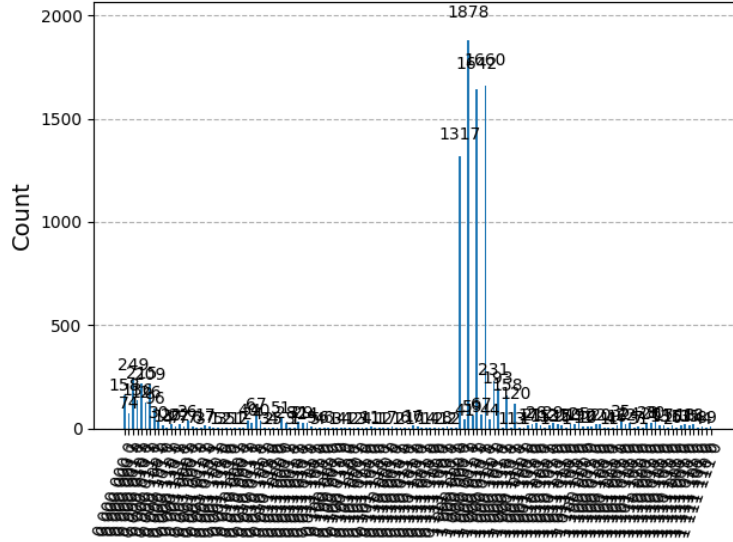


Figure 11: Measurements results obtained from running the circuit 10 000 times

By doing the post selection process we obtain a success rate of 72% and the final probability distribution of the quantum circuit is:

$$\tilde{\pi} = [0.1829 \quad 0.2609 \quad 0.2281 \quad 0.2306 \quad 0.0321 \quad 0.0268 \quad 0.0219 \quad 0.0167]$$

Comparing this sampled outcome to the true analytical stationary distribution:

$$\pi = [0.1731 \quad 0.2176 \quad 0.1992 \quad 0.2101 \quad 0.0468 \quad 0.0555 \quad 0.0606 \quad 0.0372]$$

we calculate the total variation distance (d_{TV}) between the target distribution and the simulated output:

$$d_{TV} = \frac{1}{2} \sum_i |\pi_i - \tilde{\pi}_i| \approx 0.1025$$

F Complexity: Quantum vs Classical

In the reversible case we previously analyzed, the spectral gap δ is relatively large. As a result, the degree of the fast-forwarding Chebyshev polynomial, which scales as

$$d = O\left(\frac{\log(1/\varepsilon)}{\sqrt{\delta}}\right),$$

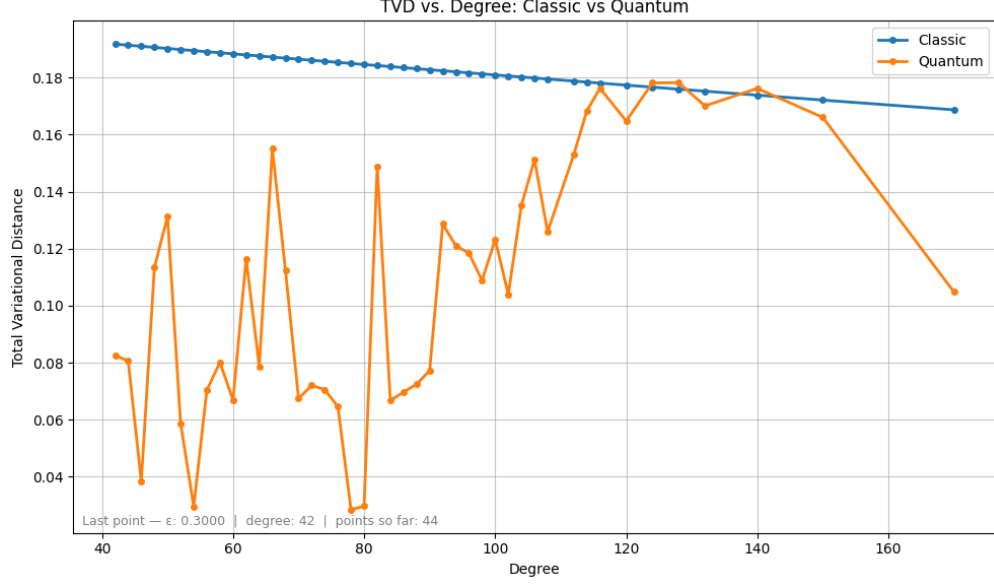


Figure 12: Classical vs Quantum: evolution of the d_{VD} as the precision (degree) increases.

is dominated by the logarithmic factor $\log(1/\varepsilon)$. Consequently, the dependence on the gap δ is negligible, and the potential quantum speedup is not clearly visible.

For the selected non-reversible chain, however, the pseudo-spectral gap $\gamma_\infty(P)$ is small, making the $1/\sqrt{\gamma_\infty(P)}$ term the dominant factor in the polynomial degree. This directly influences the circuit complexity, highlighting how the resources required scale with the spectral properties of the kernel. Therefore, analyzing this non-reversible case provides a more meaningful assessment of the quantum algorithm's efficiency.

To assess the quantum advantage, we compared the number of steps required to reach a target total variation distance (d_{TV}) from the stationary distribution. In the classical case, this corresponds to the number of applications of the Markov kernel, i.e., the degree of P^k . In the quantum case, it corresponds to the degree of the Chebyshev polynomial used in the GQSVT procedure, since $\tau_{rev} = 1$.

The results are shown in Figure 12. For a fixed degree, the quantum procedure always reaches a smaller total variation distance than the classical evolution. This indicates that, for the same effective degree, the polynomial transformation brings the output distribution closer to the stationary distribution than the corresponding number of classical Markov-chain steps.

The classical curve has the expected behavior: the total variation distance decreases smoothly as the degree increases. This reflects the progressive convergence of P^k toward the stationary distribution.

The quantum curve, instead, is less regular. Although it remains below the classical curve, it does not decrease monotonically with the degree. This behavior is not fully understood yet. One possible explanation is that the quantum estimates are affected by sampling noise and by the postselection step, which reduces the number of successful samples used to reconstruct the output distribution. To mitigate this effect, we increased the number of shots to 100000. This produced only a slight improvement, suggesting that sampling noise is not the only

source of the observed fluctuations.

Rank	Epsilon	Quantum TVD	Quantum degree	Classical degree
1	0.061101	0.028447	78	1950
2	0.178296	0.029483	54	1914
3	0.056593	0.029688	80	1908

Table 1: Best quantum total variation distances and the classical degrees required to match them.

The comparison is even clearer in Table 1. The best quantum values of the total variation distance are obtained at relatively small polynomial degrees. For example, the smallest observed quantum distance is approximately 0.028447, obtained with degree 78. To reach the same total variation distance classically, the Markov chain requires about 1950 steps. Similarly, quantum degrees 54 and 80 already achieve distances around 0.029483 and 0.029688, while the corresponding classical degrees needed to match these accuracies are 1914 and 1908, respectively.

This table highlights the practical gap between the two approaches: the quantum procedure reaches a given accuracy with a polynomial degree that is more than one order of magnitude smaller than the number of classical Markov-chain steps needed to reach the same total variation distance.

We expected the total variation distance to improve as the polynomial degree increases, since higher-degree Chebyshev polynomials should approximate the desired spectral filter more accurately. However, this monotonic decrease is not observed in the numerical data. A possible reason is numerical instability in the GQSP parameter-finding procedure. For high-degree polynomial transformations, the required rotation angles can become very small, and the synthesis of the GQSP phases may become ill-conditioned. As a consequence, increasing the degree does not necessarily lead to a better implemented transformation.

In conclusion, the numerical results confirm the expected separation between the quantum and classical procedures: the quantum method reaches the same accuracy with substantially smaller degree. The main limitation is the stability of the implementation, since larger degrees do not always produce better d_{TV} . Thus, the results are consistent with a quantum advantage, while also showing that degree selection is critical for realizing it reliably.

Bibliography

- [1] Baptiste Claudon, Jean-Philip Piquemal, and Pierre Monmarché. Quantum speedup for nonreversible markov chains. *Nature Communications*, 16(1), November 2025.
- [2] Danial Motlagh and Nathan Wiebe. Generalized quantum signal processing, 2024.